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Introduction

1.1 Function Spaces and Quotient Spaces

Definition 1.1.1

Given two topological spaces X, Y we denote by Y^X the set of all continuous functions (maps) from X to Y . This set is topologised by the so-called **compact-open-topology**, viz., the topology with the subbase as the collection of all sets of the form

$$\langle K, U \rangle = \{f \in Y^X : f(K) \subset U\}$$

where $K \subset X$ is compact and $U \subset Y$ is open.

Theorem 1.1.2 ▶ Exponential correspondence

Let X be a locally compact Hausdorff space and Y, Z be any two topological spaces.

- (a) The **evaluation map** $E : Y^X \times X \rightarrow Y$ given by $E(f, x) = f(x)$ is continuous.
- (b) A function $g : Z \rightarrow Y^X$ is continuous iff the composite $E \circ (g \times \text{Id}_X) : Z \times X \rightarrow Y$ is continuous.
- (c) If Z is also locally compact and Hausdorff, then the function

$$\psi : (Y^X)^Z \rightarrow Y^{Z \times X}$$

defined by

$$\psi(g) = E \circ (g \times \text{Id}_X)$$

is a homeomorphism.

Note.

- (a) 是说, 我们可以把 $X \rightarrow Y$ 的连续性像讲废话一样, 特殊地多复制一份放到左边, 变成 $Y^X \times X \rightarrow Y$ 的连续性. 也就是说如果 $X \rightarrow Y$ 是连续的, 我们可以预先准备一个 X 中的一个东西, 然后让 Y^X 中的家伙把 X 里面的东西给吃了, 这个操作也是连续的.
- (b) 我们发现抽象的紧开拓扑的连续性可以和具体乘积拓扑的连续性密切联系. 形象地说, 我们可以把右边的指数拽成左边的乘积. 虽然 $Z \times X \rightarrow Y$ 上的连续性, 体现出来地更破碎一点 (关于每个 (x_0, k) 附近的“块状邻域”, 而不是 $Z \rightarrow Y^X$ 连续所需要的像是包裹了 $\{x_0\} \times K$ 一样的长条邻域.). 但是由于 K 的紧性, 我们可以将“块状邻域”拼成“长条状”的.

Proof sketch.

(c): 证明映射连续, 可以利用 (b) 把指数上的东西不断地拽到左边来, 直到右边没有指数了. 发现得到的东西恰好是“依次预先准备左边要的贡品, 然后让一个家伙依次把这些贡品吃进去, 直到没有东西可以吃了”, 这个操作根据我们 (a) 中的 note 是连续的.

Proof. (a) Take any open subset U of Y . We need to show that $E^{-1}(U)$ is open. For which, we take $(f, x_0) \in E^{-1}(U)$. Since X is locally compact, there exists a compact neighbourhood K^1 of x_0 contained in $f^{-1}(U)$, that is $f \in \langle K, U \rangle$. Then for each $(g, x_1) \in \langle K, U \rangle \times K$, we have

$$E(g, x_1) = g(x_1) \in U$$

which implies that $(f, x_0) \in \langle K, U \rangle \times K \subseteq E^{-1}(U)$. Thus E is continuous.

(b) From (a) we only need to show that if $E \circ (g \times \text{Id}_X)$ is continuous, then so does g . For which take any subbasis $\langle K, U \rangle$ of Y^X , we only need to show that $g^{-1}\langle K, U \rangle$ is open. Take any $z_0 \in g^{-1}\langle K, U \rangle$, then $g(z_0) \in \langle K, U \rangle$. For each $k \in K$, we have

$$E \circ (g \times \text{Id}_X)(z_0, k) = g(z_0)(k) \in U$$

Since $E \circ (g \times \text{Id}_X)$ is continuous, there exists open neighbourhood V_k of z_0 and the open neighbourhood W_k of k such that

$$E \circ (g \times \text{Id}_X)(V_k \times W_k) \subseteq U$$

Since K is compact, we can pass on to a finite open cover $K = \bigcup_{i=1}^m W_{k_i}$,

¹That is, a compact set K such that there exists open set V , $x \in V \subseteq K$

and set $V := \bigcap_{i=1}^m V_{k_i}$. Then

$$g(V)(K) = E \circ (g \times \text{Id}_X)(V \times K) \subseteq U \implies g(V) \subseteq \langle K, U \rangle$$

Thus g is continuous.

(c) Note that from (b) it follows that ψ is well-defined and a bijection. Applying (b) with $(Y^X)^Z$ in place of Z and $Y^{Z \times X}$ place of Y^X . Then the continuity of ψ is equivalent to that of $E \circ (\psi \times \text{Id}_{Z \times Y}) : (Y^X)^Z \times (Z \times X) \rightarrow Y$ given by

$$(\lambda, z, x) \mapsto \lambda(z)(x)$$

This latter map is the composite of the two maps

$$\left((Y^X)^Z \times Z \right) \times X \rightarrow Y^X \times X \rightarrow Y$$

The continuity of this two maps follows from (a).

The continuity of ψ^{-1} is equivalent to that of $E \circ (\psi^{-1} \times \text{Id}_Z) : (Y^{Z \times X} \times Z) \rightarrow Y^X$, and is equivalent to that of $E \circ (E \circ (\psi^{-1} \times \text{Id}_Z) \times \text{Id}_X) : (Y^{Z \times X} \times Z) \times X \rightarrow Y$, given by

$$(\mu, z, x) \mapsto \mu(z, x)$$

This latter map is the evaluation map

$$Y^{Z \times X} \times (Z \times X) \rightarrow Y$$

$$E(\mu, (z, x)) = \mu(z, x)$$

which is continuous from (a). □

1.1.1 Quotient Space

Lemma 1.1.3

The following statements are equivalent.

- (i) $U \in \tau'$ iff $q^{-1}(U) \in \tau$.
- (ii) A function $g : (Y, \tau') \rightarrow (Z, \tau'')$ is continuous iff $g \circ q$ is continuous.
- (iii) For a fixed τ , τ' is the maximal topology on Y such that q is continuous.

Definition 1.1.4

Under the above conditions, we say (Y, τ') is a **quotient space** of (X, τ) and the map q is called a **quotient map**.

Proposition 1.1.5

Y is a T_1 space iff all the fibers of q are closed subsets, irrespective of whether X is T_1 or not.

Proposition 1.1.6

If $q : X \rightarrow Y$ is a surjective open mapping given by a closed relation $R \subseteq X \times X$. Then Y is a Hausdorff space, irrespective of whether X is Hausdorff or not.

Proof. Y is Hausdorff, iff Δ_Y is closed. We know from definition that

$$(q \times q)^{-1}(\Delta_Y) = R$$

Then

$$R^c = (q \times q)^{-1}(\Delta_Y^c) \implies (q \times q)(R^c) = \Delta_Y^c$$

Since q is a open map, so dose $q \times q$, and since R^c is open, so dose Δ_Y^c . Thus Δ_Y is closed, Y is Hausdorff. $\square$

Theorem 1.1.7

Let $q : X \rightarrow Y$ be a quotient map and $A \subset X$. Then the restriction map $q_A : A \rightarrow q(A) =: B$ is a quotient map if the following conditions are satisfied:

- (a) B is the intersection of an open set and a closed set in Y .
- (b) For $C \subset B$, $q_A^{-1}(C)$ is closed in A (respectively open in A) iff $q^{-1}(C)$ is closed (respectively open) in $q^{-1}(B)$.

Proof. It is obvious that q is continuous and surjective. Let C be a closed sets in B . We need to prove that $q_A^{-1}(C)$ is closed in A . We write $C = K \cap B$, where K is a closed subset of Y . C is closed in B . Then $q^{-1}(C)$ is closed in X . We know from

$$q^{-1}(C) = q^{-1}(C) \cap q^{-1}(B)$$

that $q^{-1}(C)$ is closed in $q^{-1}(B)$. Thus it follows from (b) that $q_A^{-1}(C)$ is closed in A .

Then we suppose that $q_A^{-1}(C)$ is closed in A , then $q^{-1}(C)$ is closed in $q^{-1}(B)$. We know from (a) that there is a open set U and a closed set F such that

$$B = U \cup F$$

Then we have

$$q^{-1}(B) = q^{-1}(U) \cup q^{-1}(F)$$

Where $q^{-1}(U)$ is open, $q^{-1}(F)$ is closed. There is a closed set K' of X such that

$$q^{-1}(C) = K' \cap q^{-1}(B) = q^{-1}(U) \cap (q^{-1}(F) \cap K')$$

Let $\tilde{K} = q^{-1}(F) \cap K'$, then $\tilde{K}$ is closed in X . Then

$$q^{-1}(U \setminus C) = q^{-1}(U) \setminus q^{-1}(C) = q^{-1}(U) \setminus \tilde{K}$$

which is open in X . Thus $U \setminus C$ is open in Y .

$$(U \setminus C)^c \cap B = (U \cap C^c)^c \cap B = (U^c \cup C) \cap B = C \cap B = C$$

Where $(U \setminus C)^c$ is closed in Y . Thus C is closed in B .

The remaining part is similarly.

□

Theorem 1.1.8

If Y is a locally compact Hausdorff space, then for any quotient map $q : X \rightarrow Z$, the product $q \times \text{Id}_Y$ is a quotient map.

Proof sketch.

为了利用原本的商映射的泛性质, 我们需要把冗余的乘积 $\times Y$ 丢到右边的指数上去, 这个操作就需要我们甩过去的 Y 是局部紧的 Hausdorff 空间.

Proof. Take any map $g : Z \times Y \rightarrow W$. Let $f := g \circ (q \times \text{Id}_Y) : X \times Y \rightarrow W$. If f is continuous, define

$$\begin{aligned} \hat{f} : X &\rightarrow W^Y, \\ \hat{f}(x)(y) &= f(x, y) \end{aligned}$$

Then

$$E \circ (\hat{f} \times \text{Id}_Y) : X \times Y \rightarrow W$$

$$E \circ (\hat{f} \times \text{Id}_Y)(x, y) = \hat{f}(x)(y) = f(x, y)$$

Then it follows from Theorem 1.1.2 that $\hat{f}$ is continuous. There exists a continuous function $\tilde{f} : Z \rightarrow W^Y$, such that $\tilde{f} \circ q = \hat{f}$. Then $E \circ (\tilde{f} \times \text{Id}_Y)$ is continuous, we have for $z \in Z$, take $x \in X$ such that $z = q(x)$, then

$$E \circ (\tilde{f} \times \text{Id}_Y)(z, y) = \tilde{f}(z)(y) = \tilde{f}(q(x))(y) = \hat{f}(x)(y) = f(x, y)$$

The other side,

$$g(z, y) = g(q(x), y) = g \circ (q \times \text{Id}_Y)(x, y) = f(x, y)$$

We have

$$g = E \circ (\tilde{f} \times \text{Id}_Y)$$

is continuous. □

Corollary 1.1.9

If $q_i : X_i \rightarrow Z_i, i = 1, 2$ are quotient maps such that X_1, Z_2 or Z_1, X_2 are locally compact Hausdorff spaces, then $q_1 \times q_2$ is a quotient map.

Proof. If X_1, Z_2 are locally compact Hausdorff spaces. Then

$$q_1 \times q_2 = (q_1 \times \text{Id}_{Z_2}) \circ (\text{Id}_{X_1} \times q_2) : X_1 \times X_2 \rightarrow Z_1 \times Z_2$$

is a quotient map. □

1.2 Relative Homotopy

Definition 1.2.1

Let $A \subset X$, and $f, g : X \rightarrow Y$ be any two maps such that $f(a) = g(a), \forall a \in A$. We say f is homotopic to g **relative to** A if there is a homotopy H from f to g such that $H(a, t) = f(a), \forall a \in A$. We write this

$$f \simeq g, \text{ rel } A.$$

Definition 1.2.2

Two topological pairs $(X, A), (Y, A)$ are said to have the same homotopy type if there exist maps $f : (X, A) \rightarrow (Y, A), g : (Y, A) \rightarrow (X, A)$ such that $f_A = Id_A = g_A$ and

$$g \circ f \simeq Id_X \text{ rel } A; \quad f \circ g \simeq Id_Y \text{ rel } A.$$

Definition 1.2.3

Let X be a topological space and $A \subseteq X$ a subspace, $\iota : A \hookrightarrow X$ the inclusion map.

- By a **retraction** $r : X \rightarrow A$, we mean a map r such that $r(a) = a, \forall a \in A$, equivalently, $r \circ \iota = Id_A$.
- A map $f : X \rightarrow X$ which is homotopic to Id_X is called a **deformation**.
- If $r : X \rightarrow A$ is a retraction, $\iota \circ r$ is homotopic to Id_X , then we call r a **deformation retraction**, and A a **deformation retract (DR)** of X .
- If $r : X \rightarrow A$ is a **deformation retraction**, and the homotopy from Id_X to r is relative to A , then r is called a **strong deformation retraction (SDR)**.
- If $\iota : A \hookrightarrow X$ is a homotopy equivalence, then we say A is a **weak deformation retract (WDR)** of X .

Remark.

$$SDR \implies DR \implies WDR$$

Example 1.2.4

1. Let X be a convex space, then every point $x_0 \in X$ is a SDR of X . We define a map $H : X \times \mathbb{I} \rightarrow X$,

$$H(x, t) = (1 - t)x + tx_0$$

which is well defined since X is convex. Then

$$H : Id_X \simeq c_{x_0}, \text{ rel } \{x_0\}$$

1.3 Typical Constructions

1.3.1 The Cone

Definition 1.3.1

By a cone CX over X , we mean the quotient space of $X \times I$ obtained by identifying the subspace $X \times 0$ to a single point. (See Figure 1.18.) This point is called the **vertex** or **apex** of the cone and is denoted simply by $\star$. The space X itself can then be identified with the image of the subspace $X \times 1$ in the quotient space, and this is then referred to as the **base** of the cone.

Remark.

- One can represent of point of CX by $[x, t]$ for some $x \in X$ and $t \in \mathbb{I}$. The representation is unique if $t \neq 0$. Also, $[x, 0]$ represents the vertex $\star$.
- $([x, t], s) \rightarrow [x, st]$ defines a homotopy of the constant map $\star$ with the identify map of CX , then CX is contractible.

Definition 1.3.2

Given any map $f : X \rightarrow Y$, there is a map $C(f) : CX \rightarrow CY$ defined by

$$C(f)[x, t] = [f(x), t]$$

satisfying

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ i_X \downarrow & & \downarrow i_Y \\ CX & \xrightarrow{C(f)} & CY \end{array}$$

where $i_X : X \rightarrow X \times \{1\}$, $i_Y : Y \rightarrow Y \times \{1\}$ are the identifications.

Remark.

We always omitt the map i_X and simply write $X = X \times \{1\}$.

Proposition 1.3.3

$C(f)$ is a homeomorphism iff f is a homeomorphism.

Proof. If $C(f)$ is a homeomorphism, then $C(f)|_X : X \rightarrow C(f)(X)$ is a homeomorphism with the subspace topology, where

$$C(f)(X) = [f(X), 1] = [Y, 1] = Y$$

Thus $C(f)|_X : X \rightarrow Y$ is a homeomorphism.

Conversely, if f is a homeomorphism, then f, f^{-1} give rise to continuous maps $C(f)$ and $C(f^{-1})$. Note that

$$C(f) \circ C(f^{-1})([y, t]) = [y, t], \quad C(f^{-1}) \circ C(f)([x, t]) = [x, t]$$

We know $C(f)$ is a homeomorphism. □

Theorem 1.3.4

Any topological space X is contractible iff it is a retract of the cone CX .

Note.

我们可以把 X 在时间 $\mathbb{I}$ 上的状态展开在 $X \times \mathbb{I}$ 上面.

Proof. Let $q : X \times \mathbb{I} \rightarrow CX$ denote the quotient map. Given a homotopy $H : X \times \mathbb{I} \rightarrow X$, such that $H(x, 0) = x_0, H(x, 1) = x$. There is a continuous map $r : CX \rightarrow X$ such that $H = r \circ q$. Clearly $r(x, 1) = x$ and since we have identified X with $X \times 1$, this means $X \times 1 \subseteq CX$ is a retract of CX .

Conversely, given any retraction $r : CX \rightarrow X$, the formula $H(x, t) = r([x, t])$ gives a homotopy of the constant map with the identity map of X and hence X is contractible. □

Theorem 1.3.5

Let X, Y be topological spaces. Then $f : X \rightarrow Y$ is null homotopic, if and only if f can be extend to a map $\tilde{f} : CX \rightarrow Y$

Proof sketch.

$\tilde{f}$ 就是同伦映射 $H : X \times \mathbb{I} \rightarrow Y$ 在商空间上的诱导.

Proof. If f is a null homotopic, then there exists $H : X \times \mathbb{I} \rightarrow Y$, $H(x, 1) = f(x), H(x, 0) = c$. Then H induces a map on quotient $\tilde{f} : CX \rightarrow Y$, such that $\tilde{f}(x, 1) = f(x)$.

Conversely, if f can be extend to $\bar{f} : CX \rightarrow Y$. Then we define

$$H(x, t) = \bar{f}([x, t])$$

We have

$$H(x, 0) = \bar{f}(\star), \quad H(x, 1) = \bar{f}([x, 1]) = f(x)$$

□

Example 1.3.6

The cone over S^{n-1} is homeomorphic to $\mathbb{D}^n$. Consider the map $P : S^{n-1} \times \mathbb{I} \rightarrow \mathbb{D}^n$ given by $(x, t) \mapsto tx$. The map is clearly a continuous bijection and one-to-one except that all points of $\bar{P} : CS^{n-1} \rightarrow \mathbb{D}^n$. Since the domain is compact and the range is Hausdorff, $\bar{P}$ is a homeomorphism.

Theorem 1.3.7

Let X be any topological space and $f : S^{n-1} \rightarrow X$ be any map. The following statements are equivalent:

1. $f : S^{n-1} \rightarrow X$ is null homotopic;
2. f can be extended to a map $\bar{f} : \mathbb{D}^n \rightarrow X$;
3. f is null homotopic relative to any given point $p \in S^{n-1}$.

Proof. The equivalence of 1.2. is immediate.

2. $\implies$ 3. Fix $p \in S^{n-1} \subseteq \mathbb{D}^n$. Let $F : \mathbb{D}^n \times \mathbb{I} \rightarrow \mathbb{D}^n$,

$$F(x, t) = (1 - t)x + tp$$

Then $\bar{f} \circ F|_{S^{n-1} \times \mathbb{I}} : S^{n-1} \times \mathbb{I} \rightarrow X$ is a homotopy relative to p from f to c_p .

3. $\implies$ 1. is obvious. □

1.3.2 The Adjunction Spaces

Definition 1.3.8 ▶ Adjunction space

Let X be a closed subspace of a space Z and $f : X \rightarrow Y$ be a continuous map. The **adjunction space** $Z \cup_f Y =: A_f$ of f is defined to be the quotient of the disjoint union $Y \sqcup Z$ by the identification $x \sim f(x)$ for all $x \in X$. This is also called the **space obtained by attaching Z to Y**

via the map f . Note that Y can be identified with a closed subspace of A_f via $j : y \mapsto [y]$. If the image of f is closed in Y , then the image of Z is closed in A_f . Also, if f is injective then Z can be identified with its image in A_f .

Proof. Let $q : Y \sqcup Z \rightarrow A_f$ be the quotient map. Then

$$q^{-1}(j(Y)) = Y \sqcup X$$

□

Basic Topology

2.1 度量空间

Definition 2.1.1

令 $X \neq \emptyset$ 是一个非空集合. X 上的一个度量, 是指一个映射

$$\begin{aligned} d : X \times X &\rightarrow \mathbb{R} \\ (x, y) &\mapsto d(x, y) \end{aligned}$$

使得对于任意的 $x, y, z \in X$, 有

1. $d(x, y) \geq 0$, 且 $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) \leq d(x, y) + d(y, z)$

Example 2.1.2

$X \neq \emptyset$, 定义 $d : X \times X \rightarrow \mathbb{R}$

$$d(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

是一个度量.

Proof. 1.2. 是显然的. 3. 当 $x = z$ 时是显然的. 若 $x \neq z$, 则要么 $x \neq y$, 要么 $z \neq y$, 从而

$$d(x, z) = 1 \leq d(x, y) + d(y, z)$$

□

2.2 Exercises

Problem 2.2.1 ► Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphic to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明: 1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 2.2.2

X is compact, Y is Hausdorff . If $\exists f : X \rightarrow Y$ is a continuous bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$. Since K is compact, $\text{dist}(K, 0) < \infty$. If $x \neq 0$, then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\text{dist}(K, 0) \cdot \|x\|}$. Then $\left\| \frac{x}{\varepsilon_0} \right\| > \text{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction. For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

, which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx , we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle

inequality, we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex, we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta} \left((N(x_2) + \delta) \frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta) \frac{x_1}{N(x_1) + \delta} \right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \text{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x) \frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continuous, only need to show it is continuous at 0. In fact, $\|\lim_{x \rightarrow 0} \varphi(x)\| = \lim_{x \rightarrow 0} \|\varphi(x)\| = \lim_{x \rightarrow 0} N(x) = 0$ since N is a norm. We define $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)} x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \text{Id}$. The same skills can be used to show ψ is continuous at 0 as well. Finally, by the lemma above, X, Y are homeomorphic.

□

Problem 2.2.3

Let X, X' be topological spaces and $x \in X, x' \in X'$ be base points. Prove that the wedge sum $X \vee X'$ is homeomorphic to the subspace:

$$(X \times \{x'\}) \cup (\{x\} \times X') \subseteq X \times X'$$

Proof. $X \vee X' = (X \amalg X') / \{x \sim x'\}$. Considering map $\varphi : X \times X' \rightarrow X \vee X'$

$$\varphi(x_1, x_2) = \begin{cases} x_1, & x_2 = x' \\ x_2, & x_1 = x \end{cases}$$

□

Problem 2.2.4

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to :

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translation.

Problem 2.2.5 ► Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n -dimensional projective space $\mathbb{K}\mathbb{P}^n$ is the orbit space of $\mathbb{K}^{n+1} \setminus \{0\}$ under the action of $\mathbb{K}^*$ by rescaling.

1. Prove that $\mathbb{R}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \setminus \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{C}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$, where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.
3. Prove that $\mathbb{R}\mathbb{P}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{C}\mathbb{P}^1$ is homeomorphic to $\mathbb{S}^2$.

Proof. 1. Define

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n, \quad \pi(x) = [x]$$

and define

$$f : \mathbb{S}^n \rightarrow \mathbb{RP}^n = \pi|_{\mathbb{S}^n}$$

Then f is continuous. Take any $p \in \mathbb{RP}^n$, there exists $v \in \mathbb{R}^{n+1} \setminus \{0\}$, such that $p = [v]$. Let $x = \frac{v}{\|v\|}$, then $x \in \mathbb{S}^n$,

$$f(x) = [x] = [v] = p$$

thus f is surjective. If

$$f(x) = f(y)$$

, then $[x] = [y]$, there exists $\lambda \in \mathbb{R}^*$, such that $x = \lambda y$. Since

$$\|x\| = \|\lambda y\| = |\lambda| \|y\| \implies |\lambda| = 1$$

, $\lambda = \pm 1$, $x = y$ or $x = -y$. Thus $\text{Fiber}_p(f) = \text{Orbit}_x(\{\pm 1\})$. The universal property of quotient topology gives a continuous bijection

$$\tilde{f} : \mathbb{S}^n \setminus \{\pm 1\} \rightarrow \mathbb{RP}^n$$

Since $\mathbb{S}^n \setminus \{\pm 1\}$ is compact, $\mathbb{RP}^n$ is Hausdorff, $\tilde{f}$ is a homeomorphism.

2. Similarly, let $g : \mathbb{S}^{2n+1} \rightarrow \mathbb{CP}^n$, $g(z) = [z]$. Take any $p = [w] \in \mathbb{CP}^n$, where $w \in \mathbb{C}^{n+1} \setminus \{0\}$. Let $z = \frac{w}{\|w\|}$. Then

$$g(z) = [z] = [w] = p$$

g is surjective. If

$$g(z_1) = g(z_2)$$

, then there exists $\lambda \in \mathbb{C}^*$, such that $z_1 = \lambda z_2$, then

$$\|z_1\| = |\lambda| \|z_2\| \implies |\lambda| = 1$$

which gives that $\lambda \in \mathbb{S}^1$. Thus $\text{Fiber}_p(f) \simeq \text{Orbit}_x(\mathbb{S}^1)$. Thus

$$\mathbb{CP}^n \simeq \mathbb{S}^{2n+1}/\mathbb{S}^1$$

3. (a) Only need to show that $\mathbb{S}^1/\{\pm 1\} \simeq \mathbb{S}^1$,

$$h(z) = z^2$$

induces a continuous bijection $\tilde{h} : \mathbb{S}^1 \setminus \{\pm 1\} \rightarrow \mathbb{S}^1$. $\mathbb{S}^1 \setminus \{\pm 1\}$ is compact, $\mathbb{S}^1$ is Hausdorff, $\tilde{h}$ is a homeomorphism.

(b) Since $\mathbb{S}^3/\mathbb{S}^1 \simeq \mathbb{S}^2$.

□

Problem 2.2.6

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$.
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translations.

Proof. 1. Let

$$f(x, y) := (e^{2\pi xi}, e^{2\pi yi})$$

Then

$$f(x, 0) = (e^{2\pi ix}, 1) = f(x, 1)$$

$$f(0, y) = (1, e^{2\pi yi}) = f(1, y)$$

Thus f induces a map $\bar{f}$ on $\mathbb{T}$ such that $\bar{f}([(x, y)]) = (e^{2\pi xi}, e^{2\pi yi})$, which is a continuous function. It is not hard to see that $\bar{f} : \mathbb{T} \rightarrow \mathbb{S}^1 \times \mathbb{S}^1$ is a bijection. As a continuous image of $[0, 1]^2$, $\mathbb{T}$ is compact. Take any closed subset V of $\mathbb{T}$, V is compact. $\bar{f}(V)$ is compact as the continuous image of V . Since $\mathbb{S}^1 \times \mathbb{S}^1$ is Hausdorff, $\bar{f}(V)$ is closed. Thus $\bar{f}$ is a continuous and closed bijection, thus is a homeomorphism.

2. Let

$$f(x, y) = (x, y)/\mathbb{Z}^2$$

Then Since $(x, 0) - (x, 1) \in \mathbb{Z}^2, (0, y) - (1, y) \in \mathbb{Z}^2$. f induces a continuous map $\bar{f}$ on from $\mathbb{T}$ to $\mathbb{S}^1 \times \mathbb{S}^1$. It is clear that f maps onto $\mathbb{R}^2/\mathbb{Z}^2$. If $f(x_1, y_1) = f(x_2, y_2)$, then

$$(x_1 - x_2, y_1 - y_2) \in \mathbb{Z}^2 \implies x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\}, \text{ and } y_1 = y_2 \text{ or } \{y_1, y_2\} = \{0, 1\}$$

thus $(x_1, y_1) \sim (x_2, y_2)$. Again since $\mathbb{T}$ is compact, $\mathbb{R}^2/\mathbb{Z}^2$ is Hausdorff, $\bar{f}$ is a homomorphism. □

Problem 2.2.7

Consider the orthogonal group $O_{n-1}(\mathbb{R})$ as the subgroup of $O_n(\mathbb{R})$ of matrices of the form:

$$\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in O_n(\mathbb{R}), A \in O_{n-1}(\mathbb{R}).$$

Let $O_{n-1}(\mathbb{R})$ act on $O_n(\mathbb{R})$ by left multiplication. Prove that the orbit space $O_n(\mathbb{R})/O_{n-1}(\mathbb{R})$ is homeomorphic to $\mathbb{S}^{n-1}$.

Proof sketch.

将 $O_n(\mathbb{R})$ 刻画为 $\mathbb{S}^{n-1}$ 上的群作用. 此时 $A \sim \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}$ 就是北极点的稳定子群. 一但作用是传递的, 由流形上的轨道-稳定化子定理, 就有

$$O_n(\mathbb{R})/O_{n-1}(\mathbb{R}) \simeq \mathbb{S}^{n-1}$$

Fundamental Group

3.1 Basic Constructions

Definition 3.1.1 ► Product path

Let X be a topological space. Let $\omega, \tau : \mathbb{I} \rightarrow X$ be two paths such that the terminal point $\omega(1)$ of ω is the same as the initial point $\tau(0)$ of τ . We then define $\omega * \tau : \mathbb{I} \rightarrow X$ by the formula

$$\omega * \tau(t) = \begin{cases} \omega(2t), & 0 \leq t \leq 1/2, \\ \tau(2t - 1), & 1/2 \leq t \leq 1. \end{cases} \quad (3.1)$$

Lemma 3.1.2

If $\omega_1 \simeq \omega_2$ and $\tau_1 \simeq \tau_2$ and $\omega_1 * \tau_1$ is defined then $\omega_2 * \tau_2$ is defined and we have, $\omega_1 * \tau_1 \simeq \omega_2 * \tau_2$.

Definition 3.1.3

We can use the standard parameterisation of any closed interval $[a, b]$ by the interval $[0, 1]$ and think of a map $\gamma : [a, b] \rightarrow X$ as a path in X , viz.,

$$\hat{\gamma}(t) := \gamma((b - a)t + a).$$

Theorem 3.1.4 ► Invariance under subdivision

Let $\gamma : [0, 1] \rightarrow X$ be a path, and $0 < t_1 < \dots < t_n < 1$. Then γ is path homotopic to $\widehat{\gamma|_{[0, t_1]}} * \dots * \widehat{\gamma|_{[t_n, 1]}}$.

Definition 3.1.5 ► inverse path

For a path f from x_0 to x_1 , the **inver path** $\bar{f}$ from x_1 back to x_0 is defined by

$$\bar{f}(s) = f(1 - s)$$

Proposition 3.1.6

Let f is a path from x_0 to x_1 , $\bar{f}$ is the inverse path. Then $f * \bar{f}$ and $\bar{f} * f$ are null homotopy.

Proof sketch.

定义截断路径 $f_t(s)$, 先沿着路径走 $(1-t)$ 时间, 走到 $f(1-t)$, 再在 $f(1-t)$ 处停留 t 时间.

Proof. Define

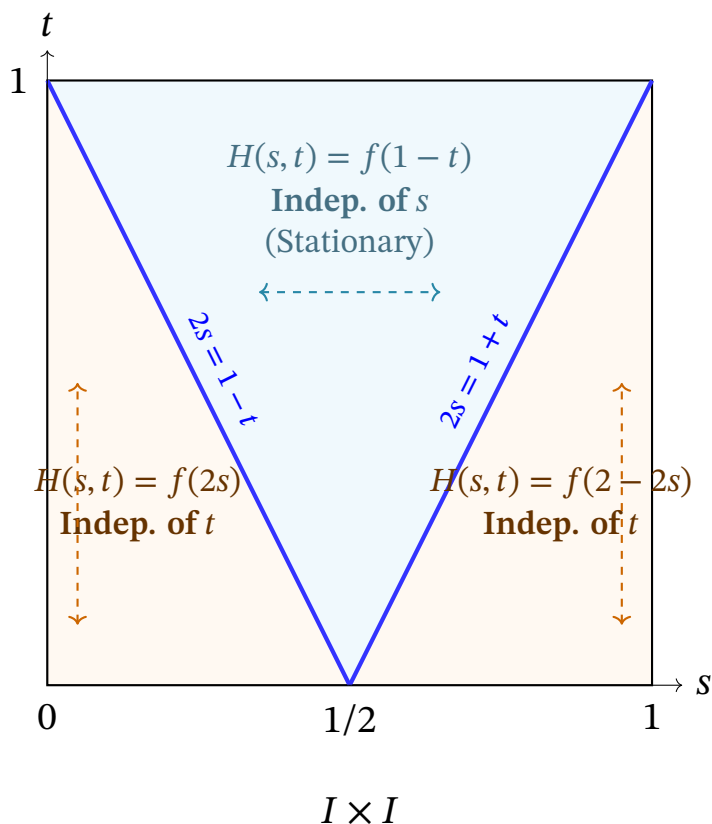
$$f_t(s) = \begin{cases} f(s), & 0 \leq s \leq 1-t \\ f(1-t), & 1-t \leq s \leq 1 \end{cases}$$

Let $g_t = \overline{f_t}$, we have

$$g_t(s) = f_t(1-s) = \begin{cases} f(1-t), & 0 \leq s \leq t \\ f(1-s), & t \leq s \leq 1 \end{cases}$$

Define $h_t = f_t * g_t$, then define $H : I \times I \rightarrow X$, $H(s, t) = h_t(s)$.

$$\begin{aligned} H(s, t) = h_t(s) &= \begin{cases} f_t(2s), & 0 \leq s \leq \frac{1}{2} \\ g_t(2s-1), & \frac{1}{2} \leq s \leq 1 \end{cases} \\ &= \begin{cases} f(2s), & 0 \leq 2s \leq 1-t \\ f(1-t), & 1-t \leq 2s \leq 1 \\ f(1-t), & 0 \leq 2s-1 \leq t \\ f(2-2s), & t \leq 2s-1 \leq 1 \end{cases} \\ &= \begin{cases} f(2s), & 0 \leq s \leq \frac{1-t}{2} \\ f(1-t), & \frac{1-t}{2} \leq s \leq \frac{t+1}{2} \\ f(2-2s), & \frac{t+1}{2} \leq s \leq 1 \end{cases} \end{aligned}$$



H is continuous on the cyan region and the two yellows region, thus is continuous. $\square$

Definition 3.1.7 ► The Fundamental Group

Let X be any topological space and $x_0 \in X$. By the above two lemmas it follows that the set $\pi_1(X, x_0)$ consisting of all path-homotopy classes of loops in X based at x_0 forms a group under the path composition. We call this group *the fundamental group* of X at x_0 .

3.2 The Fundamental Group of the Circle

Theorem 3.2.1

$\pi_1(S^1)$ is an infinite cyclic group generated by the homotopy class of the loop $\omega(s) = (\cos 2\pi s, \sin 2\pi s)$ based at $(1, 0)$.

Applications

Lemma 3.2.2

S^{n-1} is not the retract of the $\mathbb{D}^n$, that is there is no continuous map $r : \mathbb{D}^n \rightarrow S^{n-1}$.

Proof sketch.

- 借助锥构造: 收缩是恒等的延拓, $\mathbb{D}^n$ 是 S^{n-1} 锥, 能延拓到锥当且仅当是零伦. 由此导出恒等映射是零伦.

Proof. If there is such a map r , then it is an extension of $\text{Id}_{S^{n-1}}$, then $\text{Id}_{S^{n-1}}$ is a null homotopy from Theorem 1.3.7. But it follows that $\text{Id}_{\pi_{n-1}(S^{n-1})}$ is the zero map, which is a contradiction to $\pi_{n-1}(S^{n-1}) = \mathbb{Z}$ $\square$

Proof sketch.

函子性: 若收缩映射存在, 考虑

$$\pi_1(S^1) \xrightarrow{i_*} \pi_1(D^2) \xrightarrow{r_*} \pi_1(S^1)$$

而 $r_* \circ i_* = \text{Id}_{\pi_1(S^1)}$, 但是 $\pi_1(D^2) = 0$.

Theorem 3.2.3 ▶ Brouwer Fixed Point Theorem

Every continuous map $g : \mathbb{D}^n \rightarrow \mathbb{D}^n$ has a fixed point.

Proof sketch.

若没有不动点, 则每一对 $(x, g(x))$ 确定一条射线打到 S^{n-1} 上的一个点 $x \rightarrow g(x) \rightarrow S^{n-1}$, 给出连续映射 $\mathbb{D}^n \rightarrow S^{n-1}$, 与 Lemma 3.2.2 矛盾.

Lemma 3.2.4

There is no continuous map $g : S^2 \rightarrow S^1$, satisfying $g(x) = -g(-x)$.

Proof sketch.

- 存在这样的 g , 意味着我们能把 S^2 的赤道 $\eta : I \rightarrow S^2$ 通过 $g : S^2 \rightarrow S^1$ 搬到 S^1 上, $h = g \circ \eta : I \rightarrow S^1$, 它是零伦的.
- $g(x) = -g(-x)$ 导出 $h\left(s + \frac{1}{2}\right) = -h(s)$, 也就是说 $h\left(s + \frac{1}{2}\right)$ 和 $h(s)$

在 S^1 上差了半圈.

- h 提升到 $\tilde{h}$, $\tilde{h}\left(s + \frac{1}{2}\right)$ 和 $\tilde{h}(s)$ 差了个半整数.
- $\tilde{h}(0)$ 和 $\tilde{h}(1)$ 只能差奇数
- h 在 $\pi_1(S^1)$ 中对应到一个奇数, 与 h 零伦矛盾.

Theorem 3.2.5

For every continuous map $f : S^2 \rightarrow \mathbb{R}^2$ there exists a pair of antipodal points x and $-x$ in S^2 with $f(x) = f(-x)$.

Proof sketch.

若不存在, 则 $x \mapsto (f(x) - f(-x))$ 非退化, 这是一个奇映射. 我们可以将它归一化到 S^1 上, 这就与上面的 Lemma 矛盾.

3.3 Induced homomorphisms

Definition 3.3.1

Given a map $f : (X, a) \rightarrow (Y, b)$, we have a homomorphism $f_{\#} : \pi_1(X, a) \rightarrow \pi_1(Y, b)$ defined by

$$f_{\#}[\omega] = [f \circ \omega]. \quad (3.2)$$

Moreover, we have the following two properties which are verified directly.

1. For any space X , if $Id : X \rightarrow X$ denotes the identity map then the induced homomorphism on the fundamental groups is the identity homomorphism, i.e., $Id_{\#} = Id$.
2. Given maps $f : (X, a) \rightarrow (Y, b)$ and $g : (Y, b) \rightarrow (Z, c)$ we have $(g \circ f)_{\#} = g_{\#} \circ f_{\#}$.

Proof. This is well defined for if $\omega \simeq \tau$ then $f \circ \omega \simeq f \circ \tau$. Since $f \circ (\omega * \tau) = (f \circ \omega) * (f \circ \tau)$, it follows that $f_{\#}$ is a homomorphism. $\square$

Theorem 3.3.2

Let τ be a path in a space X with a, b as its initial and terminal points, respectively. Then the assignment $[\omega] \mapsto [\bar{\tau} * \omega * \tau]$ defines an isomor-

phism $h_{[\tau]} : \pi_1(X, a) \rightarrow \pi_1(X, b)$.

Proof. $h_{[\tau]}$ is a homomorphism since

$$h_{[\tau]}[\omega_1 * \omega_2] = [\bar{\tau} * \omega_1 * \omega_2 * \tau] = [\bar{\tau} * \omega_1 * \tau * \bar{\tau} * \omega_2 * \tau] = h_{[\tau]}[\omega_1] \cdot h_{[\tau]}[\omega_2]$$

Furthermore

$$h_{[\tau]} \circ h_{[\bar{\tau}]}([\omega]) = [\tau * \bar{\tau} * \omega * \tau * \bar{\tau}] = [\omega]$$

Thus

$$h_{[\tau]} \circ h_{[\bar{\tau}]} = \text{Id}_{\pi_1(X, b)}$$

Similarly

$$h_{[\bar{\tau}]} \circ h_{[\tau]} = \text{Id}_{\pi_1(X, a)}$$

□

Proposition 3.3.3

If a space X retracts onto a subspace A , then the homomorphism

$$i_* : \pi_1(A, x_0) \rightarrow \pi_1(X, x_0)$$

induced by the inclusion $i : A \hookrightarrow X$ is injective. If A is a deformation retract of X , then i_* is an isomorphism.

Proposition 3.3.4

For a space X , the following three conditions are equivalent:

- (a) Every map $S^1 \rightarrow X$ is homotopic to a constant map, with image a point.
- (b) Every map $S^1 \rightarrow X$ extends to a map $D^2 \rightarrow X$.
- (c) $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Proof. The equivalence of (a), (b) has been shown in ??

□

Proposition 3.3.5

A space X is simply-connected iff all maps $S^1 \rightarrow X$ are homotopic.

Applications

Lemma 3.3.6

If a space X is the union of a collection of path-connected open sets A_α each containing the basepoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path-connected, then every loop in X at x_0 is homotopic to a product of loops each of which is contained in a single A_α .

Proof sketch.

- 给定一个 loop $f : I \rightarrow X$.
- 取 I 的一个覆盖 $\{f^{-1}(A_\alpha)\}$ 的 Lebesgue 数, 给出 I 的一个分划 $0 = s_0 < s_1 < \dots < s_m = 1$, 每一段完整地落在某个 $f^{-1}(A_\alpha)$ 中.
- 在每个分界点 s_i 处, 到 x_0 折返一次, 切出一段回路.

Proposition 3.3.7

$\pi_1(S^n) = 0$, if $n \geq 2$.

Proof sketch.

基点取成赤道上一点, 分别取南北半球多一点的片构成覆盖, 这两个片都是可缩的, 故每个道路拆成两个可缩的道路, 是零伦.

Corollary 3.3.8

$\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.

Proof sketch.

- 设 φ 是同胚, 则 $\mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^n \setminus \{\varphi(0)\}$ 同胚.
- $\mathbb{R}^n \setminus \{\varphi(0)\}$ 平移到 $\mathbb{R}^n \setminus \{0\}$, 形变收缩到 S^{n-1} .
- $\pi_1(S^{n-1}) = 0$ 若 $n \geq 2$.
- $\mathbb{R}^2 \setminus \{0\}$ 形变收缩到 S^1 , $\pi_1(S^1) \simeq \mathbb{Z}$.
- $\pi_1(S^{n-1}) \not\simeq \pi_1(S^1)$

3.4 Van Kampen's Theorem

Theorem 3.4.1

If X is the union of path-connected open sets A_α each containing the basepoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path-connected, then the homomorphism $\Phi : *_{\alpha} \pi_1(A_\alpha) \rightarrow \pi_1(X)$ is surjective. If in addition each intersection $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, then the kernel of Φ is the normal subgroup N generated by all elements of the form $i_{\alpha\beta}(\omega)i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$, and hence Φ induces an isomorphism $\pi_1(X) \approx *_{\alpha} \pi_1(A_\alpha)/N$.

Note.

我们可以把 X 上的一个圈在内部画线, 使得它们分成 A_α 上的一些小圈, 通过在分割线上“折返”, 将大圈路径分成两个小圈的拼接. 由于在自由基的语境下, 折返是在两个不同空间上进行的, 但是对于 X 确实一个空间, 所以我们要在自由积的语境下把折返的元素 $i_{\alpha\beta}(\omega)i_{\beta\alpha}(\omega)^{-1}$ 模掉.

Exercise 3.4.2

Let X_α be a collection of path-connected spaces, we consider the wedge sum $\vee_{\alpha} X_\alpha$ with the basepoints $x_\alpha \in X_\alpha$. If each x_α is a deformation retract of an open subset U_α of X_α , then

$$\pi_1(\vee_{\alpha} X_\alpha) \simeq *_{\alpha} \pi_1(X_\alpha)$$

Proof. We have X_α is a deformation retract of its open neighborhood $A_\alpha = X_\alpha \vee_{\beta \neq \alpha} U_\beta$. The intersection of two of more A_α 's is $\vee_{\alpha} U_\alpha$, which is a deformation retract to the basepoint. Then the conclusion follows from Van Kampen's Theorem. $\square$

3.5 Covering Spaces

Definition 3.5.1

Let $p : \bar{X} \rightarrow X$ be a surjective map. We say an open subset V of X is

evenly covered by p , if $p^{-1}(V)$ is a disjoint union of open subsets of $\bar{X}$:

$$p^{-1}(V) = \coprod_i U_i$$

where, each U_i is mapped homeomorphically onto V by p . In this case, we shall refer to each U_i as a *sheet for p over V* . If the space X can be covered by open subsets which are evenly covered by p , then we say that, p is a *covering projection*; the space $\bar{X}$ is called a *covering space* of X .

3.6 Exercises

Covering Space

4.1 Basic Definitions

Definition 4.1.1 ► Covering Space

Let $p : \bar{X} \rightarrow X$ be a surjective map.

- We say an open subset V of X is **evenly covered** by p , if $p^{-1}(V)$ is a disjoint union of open subsets of $\bar{X}$:

$$p^{-1}(V) = \coprod_i U_i$$

where, each U_i is mapped homeomorphically onto V by p .

- In this case, we shall refer to each U_i as a **sheet** for p over V .
- If the space X can be covered by open subsets which are evenly covered by p , then we say that p is a **covering projection**; the space $\bar{X}$ is called a covering space of X .

Theorem 4.1.2

Let $p : \bar{X} \rightarrow X$ be a continuous map, where X is locally path connected.

1. The map p is a covering space iff for each component $\bar{C}$ of $\bar{X}$, the restriction map $p : p^{-1}(C) \rightarrow C$ is a covering projection.
2. If p is a covering projection then for each component $\bar{C}$ of $\bar{X}$, the map $p : \bar{C} \rightarrow p(\bar{C})$ is a covering projection and $p(\bar{C})$ is a component of X .

Remark.

Thus, we always assume that a covering space is locally path connected and connected. For the following content, unless otherwise specified, we will all adopt this assumption.

Proposition 4.1.3

Let $p : \bar{X} \rightarrow X$ be a covering projection. If X is Hausdorff then $\bar{X}$ is Hausdorff. Further, if p is finite-to-one, then show that if $\bar{X}$ is Hausdorff then X is Hausdorff.

Note.

I haven't finish the proof of the second part yet. Actually, if a covering projection is proper it is also closed. Once we have this conclusion we can then finish the proof easily.

Proof. 1. Take two different points $\bar{x}, \bar{y} \in \bar{X}$. Denote $x := p(\bar{x}), y := p(\bar{y})$.

- (a) If $x \neq y$: Since X is Hausdorff, there exists two open sets U, V such that $x \in U, y \in V, U \cap V = \emptyset$. We have $p^{-1}(U), p^{-1}(V)$ are two open subsets of $\bar{X}$ with $p^{-1}(U) \cap p^{-1}(V) = \emptyset$. Since $\bar{x} \in p^{-1}(U), \bar{y} \in p^{-1}(V)$, it shows that $\bar{X}$ is Hausdorff.
- (b) If $x = y$: Let W be a open neighbourhood of $x = y$. We write $p^{-1}(W) = \coprod_i W_i$. Since $\bar{x}, \bar{y} \in \coprod_i W_i$ and any two W_i are disjoint, there are two different W_{i_1}, W_{i_2} such that $\bar{x} \in W_{i_1}, \bar{y} \in W_{i_2}$.

The above shows that $\bar{X}$ is Hausdorff.

- 2. Take two different points $x, y \in X$. Then $p^{-1}(x), p^{-1}(y)$ are two finite sets of $\bar{X}$ with empty intersection. We write

$$p^{-1}(x) = \{\bar{x}_i\}_{i=1}^n, \quad p^{-1}(y) = \{\bar{y}_i\}_{i=1}^n$$

For each $i = 1, \dots, n$, let $U_i \ni \bar{x}_i, V_i \ni \bar{y}_i, U_i \cap V_i = \emptyset$. Then $p(\bigcup_i U_i) \cap p(\bigcup_i V_i) = \emptyset, x \in p(\bigcup_i U_i), y \in p(\bigcup_i V_i)$.

□

Basic Homology Algebra

5.1 Basic definition

Definition 5.1.1

Consider a direct sum

$$C_{\bullet} := C_* := \bigoplus_{n \in \mathbb{Z}} C_n$$

of R -modules^a. Often we call C_* a **graded-module** with its n^{th} **graded component** C_n . Members of C_n are also called **homogeneous elements** of C_* of degree n .

1. A **R -module homomorphism** $f : C_* \rightarrow C'_*$ is called a graded homomorphism if there exists d such that $f(C_r) \subseteq C'_{r+d}$ for all r . We then call d the degree of f . We shall denote $f|_{C_r}$ by f_r , and often we may simply write f itself for f_r provided that there is no confusion.
2. By a **chain complex** (C_*, ∂) of R -modules, we mean a graded R -module C_* , together with an endomorphism $\partial := \partial_* : C_* \rightarrow C_*$ of degree -1 with the property $\partial \circ \partial = 0$. The endomorphism ∂ is called the **differential** or the **boundary map** of the chain complex. Often we shall not mention the ∂ at all and merely say C_* is a chain complex.
3. If C and C_* are two chain complexes then by a **chain map** $f = f_* : C_* \rightarrow C'_*$ we mean a graded module homomorphism of degree 0 that commutes with the corresponding differentials.

^ais also a R -module

Remark.

1. The direct sum C_* is also an R -module.
2. Observe that ∂ consists of a sequence $\{\partial_n : C_n \rightarrow C_{n-1}\}$ of R -module homomorphisms such that $\partial_n \circ \partial_{n-1} = 0$ for all n .
3. f consists of a sequence $\{f_n : C_n \rightarrow C'_n\}$ of R -module homomorphisms such that $\partial'_n \circ f_n = f_{n-1} \circ \partial_n$ for all n . Expressed with an

diagram, that is

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ C'_n & \xrightarrow{\partial'_n} & C'_{n-1} \end{array}$$

Proposition 5.1.2

There is a category of chain complexes of R -modules and chain maps. We shall denote this category by $\mathbf{Ch}_R$.

Proof. The objects are chain complexes. The morphisms are chain maps. For composition of morphisms, consider two chain maps $f : C_* \rightarrow C'_*$, $g : C'_* \rightarrow C''_*$. Only need to show that the following diagram commutes

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ C'_n & \xrightarrow{\partial'_n} & C'_{n-1} \\ \downarrow f'_n & & \downarrow f'_{n-1} \\ C''_n & \xrightarrow{\partial''_n} & C''_{n-1} \end{array}$$

which is obvious. Finally, the existence of the identity map is also obvious. □

Definition 5.1.3 ▶ Direct Sum of Chain Complexes

Define the direct sum of a family of chain complexes $\{(C^\alpha, \partial^\alpha)\}_{\alpha \in \Lambda}$ as the chain complexes $(C, \partial) := (\bigoplus_\alpha C^\alpha, \bigoplus_\alpha \partial^\alpha)$, where the n^{th} graded component of C is

$$C_n = (C_n^\alpha)_{\alpha \in \Lambda},$$

∂ is defined as

$$\partial((c^\alpha)_{\alpha \in \Lambda}) := (\partial^\alpha(c^\alpha))_{\alpha \in \Lambda}$$

5.2 Exact Sequence

Definition 5.2.1 ▶ Exact Sequence

1. A sequence of R -modules

$$M' \xrightarrow{\alpha} M \xrightarrow{\beta} M''$$

is said to be **exact** at M if $\ker \beta = \text{Im } \alpha$.

2. A sequence

$$\cdots \longrightarrow M_{n-1} \longrightarrow M_n \longrightarrow M_{n+1} \longrightarrow \cdots$$

is said to be exact if it is exact at each M_n .

3. By a short exact sequence we mean an exact sequence of the form

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

Definition 5.2.2 ▶ Exact Sequence of Chain Complexes

A sequence of chain complexes and chain maps

$$0 \longrightarrow C' \xrightarrow{f} C \xrightarrow{g} C'' \longrightarrow 0$$

is said to be exact if for each n the corresponding sequence of modules

$$0 \longrightarrow C'_n \xrightarrow{f_n} C_n \xrightarrow{g_n} C''_n \longrightarrow 0$$

is exact.

Lemma 5.2.3 ▶ Snake lemma

Given a commutative diagram of R -module homomorphisms: where the two horizontal sequences are exact, there exists a R -module homomorphism

$$\begin{array}{ccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\alpha'} & N & \xrightarrow{\beta'} & N'' \end{array}$$

$\delta : \text{Ker } f'' \longrightarrow \text{Coker } f'$, called the *connecting homomorphism* such that the sequence

$$\text{Ker } f' \xrightarrow{\bar{\alpha}} \text{Ker } f \xrightarrow{\bar{\beta}} \text{Ker } f'' \xrightarrow{\delta} \text{Coker } f' \xrightarrow{\bar{\alpha}'} \text{Coker } f \xrightarrow{\bar{\beta}'} \text{Coker } f''$$

is exact. Moreover, the **connecting homomorphism** δ has the naturality properties, so that the above assignment of a ‘snake’ to the corresponding ‘six-term’ exact sequence of modules defines a covariant functor.

Note.

最关键的信息是 **connecting homomorphism** δ . 构造的过程就是借助 β 的是 surjective, α' 是 injective, 来调整使得 β 和 α' 对应的箭头是可以反转的. 事实上在应用时, 这个 connecting homomorphism 的具体构造往往比它的存在性更重要.

Proof. to be finished We need to show

1. $\bar{\alpha}, \bar{\beta}$ are well defined, where $\bar{\alpha} := \alpha|_{\ker f'}, \bar{\beta} := \beta|_{\ker f}$
2. $\text{Im } \bar{\alpha} = \ker \bar{\beta}$.
3. $\bar{\alpha}', \bar{\beta}'$ are well defined, where $\bar{\alpha}'([x]) := [\alpha'(x)], \bar{\beta}'([y]) := [\beta'(y)]$
4. $\text{Im } \bar{\alpha}' = \ker \bar{\beta}'$
5. Construct a δ .
6. $\text{Im } \bar{\beta} = \ker \delta$
7. $\text{Im } \delta = \ker \bar{\alpha}'$

We prove the above points below.

1. By commutative, $f \circ \alpha(\ker f') = \alpha' \circ f'(\ker f') = 0$, which shows that $\alpha(\ker f') \subseteq \ker f$. Thus $\bar{\alpha}$ is well defined. $\bar{\beta}$'s is similar.
- 2.

$$\text{Im } \bar{\alpha} = \alpha(\ker f') \subseteq \text{Im } \alpha = \ker \beta \implies \beta(\text{Im } \bar{\alpha}) = 0 \implies \text{Im } \bar{\alpha} \subseteq \ker \bar{\beta}$$

The other side,

$$\ker \bar{\beta} = \ker \beta \cap \ker f = \text{Im } \alpha \cap \ker f$$

Take $y \in \ker \bar{\beta}$, there exists $x \in M'$ such that $y = \alpha(x)$ and $f(y) = f(\alpha(x)) = 0$. Since $f \circ \alpha = \alpha' \circ f'$, $\alpha' \circ f'(x) = 0$. Since α' is injective, $x \in \ker f'$. Thus $y \in \text{Im } \bar{\alpha}$.

3. For $\bar{\alpha}'$, we need to show that for $x_1, x_2 \in N'$ such that $x_1 - x_2 \in \text{Im } f'$, $\alpha'(x_1) - \alpha'(x_2) \in \text{Im } f$. It is true by commutative

$$\alpha'(x_1) - \alpha'(x_2) = \alpha'(x_1 - x_2) \in \text{Im } (\alpha' \circ f') = \text{Im } (f \circ \alpha) \subseteq \text{Im } f$$

$\bar{\beta}'$'s is similar.

4. Take $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, then there exists $[x]_{\sim \text{Im } f'} \in \text{Coker } f'$, such that $\alpha'([x]_{\sim \text{Im } f'}) = [y]_{\sim \text{Im } f}$. Then $\alpha'(x) - y \in \text{Im } f$, there exists $z \in M$, such that

$$\alpha'(x) - y = f(z)$$

By commutative and exactness: $\ker \beta' = \text{Im } \alpha'$,

$$\beta'(y) = \beta'(\alpha'(x) - f(z)) = 0 - \beta' \circ f(z) = -f'' \circ \beta(z) \in \text{Im } f''$$

, which shows that $\bar{\beta}'([y]_{\sim \text{Im } f}) = [0]_{\sim \text{Im } f''}$, $\text{Im } \bar{\alpha}' \subseteq \ker \bar{\beta}'$.

Then, take $[y]_{\sim \text{Im } f} \in \ker \bar{\beta}'$. Then $y \in \ker \beta' = \text{Im } \alpha'$. There exists $x \in N'$ such that $y = \alpha'(x)$. Then $[y]_{\sim \text{Im } f} = \alpha'([x]_{\sim \text{Im } f'})$, $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, $\ker \bar{\beta}' \subseteq \text{Im } \bar{\alpha}'$.

5. Take $z \in \ker f''$. Since β is surjective, there exists $y \in M$ such that $\beta(y) = z$. Once we show that $f(y) \in \text{Im } \alpha'$, we can define

$$\delta(z) = [(\alpha')^{-1}(f(y))]_{\sim \text{Im } f'}$$

provided that this definition is well defined. By commutative

$$\beta'(f(y)) = f''(\beta(y)) = f''(z) = 0.$$

Hence $f(y) \in \ker \beta' = \text{Im } \alpha'$. Finally, to show that it is well defined, we take $z_1, z_2 \in \ker f''$ such that $\beta(y_1) = \beta(y_2) = z$. We need to show that $(\alpha')^{-1}(f(y_1 - y_2)) \in \text{Im } f'$. Since $\beta(y_1) - \beta(y_2) = 0$, $\beta(y_1 - y_2) = 0$, $y_1 - y_2 \in \ker \beta = \text{Im } \alpha$. Suppose $\alpha(x) = y_1 - y_2$. Then

$$(\alpha')^{-1}(f(y_1 - y_2)) = (\alpha')^{-1}(f\alpha(x)) = (\alpha')^{-1}(\alpha' \circ f')(x) = f'(x) \in \text{Im } f'$$

, which completes the proof. □

Corollary 5.2.4

Consider the following commutative diagram of R -modules and R -linear maps in which the two rows are exact. If f_1 and f_3 are isomorphisms then so is f_2 .

$$\begin{array}{ccccccc}
0 & \longrightarrow & M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 \longrightarrow 0 \\
& & \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 \\
0 & \longrightarrow & N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 \longrightarrow 0
\end{array}$$

Proof. By snake lemma, we have

$$0 \xrightarrow{\alpha_1} \ker f_2 \xrightarrow{\alpha_2} 0 \xrightarrow{\delta} 0 \xrightarrow{\beta_1} \operatorname{Coker} f_2 \xrightarrow{\beta_2} 0$$

is exact. Then we have

$$\begin{aligned}
0 &= \ker \alpha_2 = \operatorname{Im} \alpha_1 = 0 \\
&= \ker f_2
\end{aligned}$$

which shows that $\ker f_2 = 0$. Similarly, $\operatorname{Coker} f_2 = 0$. The above shows that f_2 is an isomorphism. $\square$

Corollary 5.2.5 ▶ Four lemma

Consider the following commutative diagram of R modules and R -linear maps in which the two rows are exact. Suppose that f_1 is surjective and f_4 is injective. Then

- (i) f_2 is injective $\implies f_3$ is injective.
- (ii) f_3 is surjective $\implies f_2$ is surjective.

$$\begin{array}{ccccccc}
M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & M_4 \\
\downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 \\
N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4
\end{array}$$

Proof sketch.

如果 f_3 被两个单射 f_2, f_4 夹在中间, 得益于 f_1 是满射, 可以将右三列保持单射性地调整为一条 snake, 从而导出 f_3 是单射.

类似地, 如果 f_2 被两个满射 f_1, f_3 夹在中间, 得益于 f_4 是单射, 左三列可以保持满射性地调整为一条 snake, 导出 f_2 是满射.

Proof. 1. We can construct a snake

$$\begin{array}{ccccccc}
 M_2 / \ker \alpha_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & \operatorname{Im} \alpha_3 & \longrightarrow & 0 \\
 \downarrow \bar{f}_2 & & \downarrow f_3 & & \downarrow f_4|_{\operatorname{Im} \alpha_3} & & \\
 0 \longrightarrow & N_2 / \ker \beta_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 &
 \end{array}$$

By snake lemma, the following sequence is exact

$$\ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3}$$

Since f_4 is injective, $f_4|_{\operatorname{Im} \alpha_3}$ is as well. Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{p} & M_2 / \ker \alpha_2 & \longrightarrow & 0 \\
 \bar{f}_1 \downarrow & & \downarrow f_2 & & \downarrow \bar{f}_2 & & \\
 0 \longrightarrow & N_1 / \ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{p} & N_2 / \ker \beta_2 &
 \end{array}$$

We get

$$0 = \ker f_2 \rightarrow \ker \bar{f}_2 \xrightarrow{\delta} \operatorname{Coker} \bar{f}_1$$

is a exact suquence, where $\operatorname{Coker} \bar{f}_1 = 0$ since f_1 is surjective and $\bar{f}_1$ is as well. It shows that $\ker \bar{f}_2 = 0$. Finally, we have the following sequence is exact

$$0 = \ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3} = 0$$

It follows that $\ker f_3 = 0$, f_3 is injective.

2. Another snake we can construct is

$$\begin{array}{ccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & \operatorname{Im} \alpha_2 = \ker \alpha_3 & \longrightarrow & 0 \\
 \downarrow \bar{f}_1 & & \downarrow f_2 & & \downarrow f_3|_{\operatorname{Im} \alpha_2} & & \\
 0 \longrightarrow & N_1 / \ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{\beta_2} & \operatorname{Im} \beta_2 = \ker \beta_3 &
 \end{array}$$

Then by snake lemma ,the following sequence is exact

$$\ker \bar{f}_1 \rightarrow \ker f_2 \rightarrow \ker f_3|_{\operatorname{Im} \alpha_2} \rightarrow \operatorname{Coker} \bar{f}_1 \rightarrow \operatorname{Coker} f_2 \rightarrow \operatorname{Coker} f_3|_{\operatorname{Im} \alpha_2}$$

Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & \text{Ker } \alpha_3 & \xrightarrow{\iota_M} & M_3 & \xrightarrow{\alpha_3} & \text{Im } \alpha_3 & \longrightarrow & 0 \\
 & & \downarrow f_3|_{\text{Im } \alpha_2} & & \downarrow f_3 & & \downarrow f_4|_{\text{Im } \alpha_3} & & \\
 0 & \longrightarrow & \text{Ker } \beta_3 & \xrightarrow{\iota_N} & N_3 & \xrightarrow{\beta_3} & \text{Im } \beta_3 & \longrightarrow & 0
 \end{array}$$

We get

$$0 = \ker f_4|_{\text{Im } \alpha_3} \xrightarrow{\delta} \text{Coker}(f_3|_{\text{Im } \alpha_2}) \rightarrow \text{Coker } f_3 = 0$$

which shows that $\text{Coker}(f_3|_{\text{Im } \alpha_2}) = 0$. Finally, we have the following sequence exact

$$0 = \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2} = 0$$

Thus $\text{Coker } f_2 = 0$, f_2 is surjective.

□

Corollary 5.2.6 ► Five lemma

In the following diagram of R -modules, the two rows are given to be exact. If f_1, f_2, f_4 and f_5 are isomorphisms then f_3 is also an isomorphism.

$$\begin{array}{ccccccccc}
 M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & M_4 & \longrightarrow & M_5 \\
 \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\
 M'_1 & \longrightarrow & M'_2 & \longrightarrow & M'_3 & \longrightarrow & M'_4 & \longrightarrow & M'_5
 \end{array}$$

Proof. By applying the Four lemma to the left four columns, we get f_3 is injective. And by applying the Four lemma to the right four columns, we get f_3 is surjective.

□

5.3 Homology

Definition 5.3.1

Given a chain complex C_* , define the **homology group** of C_* to be the graded R -module

$$H_*(C_*) := \bigoplus_{n \in \mathbb{Z}} H_n(C_*)$$

by taking

$$H_n(C_*) := \text{Ker } \partial_n / \text{Im } \partial_{n+1}, \quad \forall n \in \mathbb{Z}.$$

Proposition 5.3.2

If $f : C_* \rightarrow C'_*$ is a chain map then f induces a graded homomorphism

$$H_*(f) : H_*(C_*) \rightarrow H_*(C'_*)$$

In addition, this has the naturality property, viz.,

1. $H_*(\text{Id}) = \text{Id}$
2. If g is another chain map such that $f \circ g$ is defined, then

$$H_*(f \circ g) = H_*(f) \circ H_*(g)$$

Thus, H_* is a **covariant functor** from the category of chain complexes to the category of graded modules.

Theorem 5.3.3

The homology of a direct sum of chain complexes is isomorphic to the direct sum of the homology of chain complexes.

Proof.

$$H_n(\bigoplus_{\alpha} C_*^{\alpha}) = \text{ker}(\bigoplus_{\alpha} \partial_n^{\alpha}) / \text{Im}(\bigoplus_{\alpha} \partial_{n+1}^{\alpha})$$

$$\bigoplus_{\alpha} H_n(C_*^{\alpha}) = \bigoplus_{\alpha} (\text{ker } \partial_n^{\alpha} / \text{Im } \partial_{n+1}^{\alpha})$$

There are obvious isomorphisms :

$$\text{ker}(\bigoplus_{\alpha} \partial_n^{\alpha}) \simeq \bigoplus_{\alpha} \text{ker } \partial_n^{\alpha}, \quad \text{Im}(\bigoplus_{\alpha} \partial_{n+1}^{\alpha}) \simeq \bigoplus_{\alpha} \text{Im } \partial_{n+1}^{\alpha}$$

A standard result in algebra is that if K_α is a submodule of M_α , then

$$\frac{\bigoplus_\alpha M_\alpha}{\bigoplus_\alpha K_\alpha} \cong \bigoplus_\alpha \frac{M_\alpha}{K_\alpha}$$

. Hence , we have

$$\bigoplus_\alpha \ker \partial_n^\alpha / \bigoplus_\alpha \operatorname{Im} \partial_{n+1}^\alpha \simeq \bigoplus_\alpha (\ker \partial_n^\alpha / \operatorname{Im} \partial_{n+1}^\alpha)$$

□

Theorem 5.3.4

Given a short exact sequence of chain complexes

$$0 \rightarrow C'_* \xrightarrow{\alpha} C_* \xrightarrow{\beta} C''_* \rightarrow 0$$

there is a functorial long exact sequence of homology groups

$$\rightarrow H_n(C'_*) \xrightarrow{H_n(\alpha)} H_n(C_*) \xrightarrow{H_n(\beta)} H_n(C''_*) \xrightarrow{\delta_n} H_{n-1}(C'_*) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(C_*) \rightarrow$$

Proof sketch.

Consider the diagram

$$\begin{array}{ccccccc} C'_n / \operatorname{Im} \partial'_{n+1} & \xrightarrow{\bar{\alpha}_n} & C_n / \operatorname{Im} \partial_{n+1} & \xrightarrow{\bar{\beta}_n} & C''_n / \operatorname{Im} \partial''_{n+1} & \longrightarrow & 0 \\ \downarrow \partial'_n & & \downarrow \partial_n & & \downarrow \partial''_{n+1} & & \\ 0 \longrightarrow & \operatorname{Ker} \partial'_{n-1} & \xrightarrow{\alpha'_{n-1}} & \operatorname{Ker} \partial_{n-1} & \xrightarrow{\beta'_{n-1}} & \operatorname{Ker} \partial''_{n-1} & \end{array}$$

Singular Homology

6.1 The Construction

Definition 6.1.1

Let X be any topological space and $n \geq 0$ be an integer. By a **singular n -simplex** in X , we mean a continuous map $\sigma : |\Delta_n| \rightarrow X$. By a **singular n -chain** in X , we mean a finite sum $\sum_i n_i \sigma_i$, where $n_i \in \mathbb{Z}$ and σ_i are singular n -simplices in X . We can add any two chains in an obvious way to obtain another chain by simply following the rule

$$n\sigma + m\sigma = (n + m)\sigma \quad (4.5)$$

Then the collection of all singular n -chains in X becomes a free abelian group, with the empty sum playing the role of the zero-element. The collection of all singular n -simplices is then a basis for this group. We denote this group by $S_n(X)$.

Proposition 6.1.2 ► Functoriality

If $f : X \rightarrow Y$ is a map of topological spaces, then $\sigma \mapsto f \circ \sigma$ extends to define a graded group homomorphism $f_* : S_n(X) \rightarrow S_n(Y)$. Since $(g \circ f) \circ \sigma = g \circ (f \circ \sigma)$, it follows that the association $X \mapsto S_n(X)$ defines a functor.

Definition 6.1.3 ► Relative Chain Complex

If A is a subspace of X and σ is a singular simplex in A then it can be considered as a singular simplex in X via the inclusion map $A \hookrightarrow X$. In this way, $S_n(A)$ becomes a subgroup of $S_n(X)$. We denote the quotient group $S_n(X)/S_n(A)$ by $S_n(X, A)$.

Remark.

1. $S_n(X, A)$ is a free abelian group with the collection of all singular n -simplexes σ in X whose image is not contained in A as a basis.

2. If $A = \emptyset$, then clearly $S_n(X, A) = S_n(X)$.

Definition 6.1.4

For each integer $r \geq 0$, the **face map** or the **face operator** $F^r : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ is defined by

$$F^r(\mathbf{e}_s) = \begin{cases} \mathbf{e}_s, & \text{if } s < r, \\ \mathbf{e}_{s+1}, & \text{if } s \geq r \end{cases}$$

and extended linearly.

Remark.

1. Note that for each $n \geq r$, F^r carries Δ_{n-1} onto the $(n-1)$ -face of Δ_n opposite to the vertex $\mathbf{e}_r$, in an **order preserving** manner.
2. We shall denote each F^r restricted to any $\Delta_n, n \geq r$ also by F^r , the exact meaning being understood from the context.

Lemma 6.1.5

$$F^r \circ F^{s-1} = F^s \circ F^r \text{ if } r < s. \text{ }^a$$

^a这里修正了 shastri 书中的一个笔误, 他可能写错了 $s-1$ 的位置

Proof. Since $r < s, r \leq s-1$, we fully expand the LHS by definition

$$\begin{aligned} F^r \circ F^{s-1}(\mathbf{e}_k) &= \begin{cases} F^r(\mathbf{e}_k), & k < s-1 \\ F^r(\mathbf{e}_{k+1}), & k \geq s-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < s-1, k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+1}, & s-1 \leq k < r-1 \text{ (empty set)} \\ \mathbf{e}_{k+2}, & k \geq s-1, k \geq r-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases} \end{aligned}$$

Similarly,

$$\begin{aligned}
 F^s \circ F^r (\mathbf{e}_k) &= \begin{cases} F^s (\mathbf{e}_k), & k < r \\ F^s (\mathbf{e}_{k+1}), & k \geq r \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r, k \leq s \\ \mathbf{e}_{k+1}, & k < r, k \geq s \text{ (empty set)} \\ \mathbf{e}_{k+1}, & k \geq r, k < s-1 \\ \mathbf{e}_{k+2}, & k \geq r, k \geq s-1 \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases}
 \end{aligned}$$

□

Definition 6.1.6

For $n \geq 1$ and for any singular n -simplex σ in X , we define

$$\partial_n(\sigma) = \sum_{r=0}^n (-1)^r \sigma \circ F^r$$

and extend it linearly to obtain homomorphisms $\partial_n : S_n(X, A) \rightarrow S_{n-1}(X, A)$. We define $S_n(X, A) = (0)$ for $n \leq -1$ and $\partial_n = 0$ for $n \geq 0$.

Note.

σ is a map from Δ_n to X . $\sigma \circ F^r$ is the restriction of σ on its face opposite to $\mathbf{e}_r$. Then $\partial_n(\sigma)$ is something like a closed $(n-1)$ -surface consists of σ 's faces.

Proposition 6.1.7 ► Singular Chain Complex

$(S(X, A), \partial) = \{S_n(X, A), \partial_n\}$ is a chain complex and the association $(X, A) \rightsquigarrow (S(X, A), \partial)$ is a functor.

Proof. We shall first show that $\partial^2(\sigma) = 0$ for any singular n -simplex, $n \geq 2$:

$$\begin{aligned}
 \partial^2(\sigma) &= \partial \left(\sum_{s=0}^n (-1)^s \sigma \circ F^s \right) \\
 &= \sum_{s=0}^n (-1)^s \partial(\sigma \circ F^s) \\
 &= \sum_{s=0}^n \sum_{r=0}^{n-1} (-1)^s (-1)^r \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq r \geq s \geq 0} (-1)^{s+r} \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq s \geq r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^s \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} - \sum_{n \geq s > r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} \\
 &= 0
 \end{aligned}$$

Given a map $f : (X, A) \rightarrow (Y, B)$. Since $(f \circ \sigma) \circ F^r = f \circ (\sigma \circ F^r)$, we define $f_*(\sigma) := f \circ \sigma$ extended by linearity. Then $\partial(f_*(\sigma)) = f_*(\partial(\sigma))$ by the definition of ∂ and f_* 's linearity. The rest of the details are omitted. $\square$

Definition 6.1.8

The **relative singular homology groups** of a topological pair (X, A) are the homology groups $H_*(S_*(X, A))$ and are denoted by $H_*(X, A)$. Taking $A = \emptyset$, we get the homology groups $H_*(X)$.

Remark.

Given any commutative ring R

1. We could have formed the **free module** $S_n(X; R)$ of all the singular n -chains in X .
2. Exactly as before, we would have got a chain complex $S_*(X; R)$, sub-chain complex $S_*(A; R)$ and the quotient complex $S_*(X, A; R)$, etc.
3. The corresponding homology groups $H_*(X; R)$, etc., are then called the homology groups X with coefficients in R . Clearly they form a **graded R -module** by themselves. It is customary to drop the notation R when $R = \mathbb{Z}$.

6.2 Basic Properties

Remark.

- It is easily seen that $\{S_*(X, A), \partial\}$ is a functor from the category of topological pairs to the category of chain complexes; thus $H_*(X, A)$ and $H_*(X)$ are also functors.
- Given $f : (X, A) \rightarrow (Y, B)$, we shall denote by f_* , the map induced by f on $S_*(X, A)$ and by f_* the map induced by f on $H_*(X, A)$.

Lemma 6.2.1

There exist functorial chain homotopies

$$h : S_q(X, A) \rightarrow S_{q+1}((X, A) \times \mathbb{I}), q \geq 0,$$

,called **prism operator**, between the two inclusion induced maps $\eta_0, \eta_1 : S_*(X, A) \rightarrow S_*((X, A) \times \mathbb{I})$, where for any $t \in \mathbb{I}$, η_t is the inclusion map

$$\eta_t(x) = (x, t), \quad t \in \mathbb{I}$$

Theorem 6.2.2 ▶ Homotopy invariance

Let $f, g : (X, A) \rightarrow (Y, B)$ be **homotopic** to **each other**. Then $f_* = g_*$ on $H_*(X, A)$.

Proof. Let $G : (X, A) \times \mathbb{I} \rightarrow (Y, B)$ be a homotopy between f and g . For any $t \in \mathbb{I}$, let $\eta_t : (X, A) \rightarrow (X, A) \times \mathbb{I}$ be inclusion map $\eta_t(x) = (x, t)$. Then $G \circ \eta_0 = f$, $G \circ \eta_1 = g$. Only need to show that chain maps $(\eta_0)_*$ and $(\eta_1)_*$ are chain homotopic, thus $(\eta_0)_* = (\eta_1)_*$, $f_* = g_*$, which is implied by the lemma above. $\square$

Corollary 6.2.3

Homotopy-equivalent topological spaces have isomorphic homology groups; in particular, a contractible space has the homology groups of a point-space.

Example 6.2.4

Let $\star$ be a point space. We have

$$H_k(\star) = \begin{cases} \mathbb{Z}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

Proof. Let n be any positive integer. There is a unique singular n -simplex $\sigma : |\Delta_n| \rightarrow \{\star\}$ and hence $S_n(\star) \cong \mathbb{Z}$ for each $n \geq 0$. Observe that $\partial : S_{2n+1}(\star) \rightarrow S_{2n}(\star)$ is the zero-map because Δ_{2n+1} has an even number of $(2n)$ -faces and the terms in $\partial(\sigma)$ cancel in pairs. However, similar reasoning tells us that $\partial : S_{2n}(\star) \rightarrow S_{2n-1}(\star)$ is an isomorphism. Thus we have a chain complex

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \longrightarrow 0$$

Thus it follows that $H_0(\star) \cong \mathbb{Z}$ and $H_i(\star) = (0)$ for all $i > 0$. $\square$

Example 6.2.5

Let $\{X_j : j \in J\}$ be the set of path connected components of X so that $X = \bigsqcup_{j \in J} X_j$. Then

$$H_*(X) = \bigoplus_{j \in J} H_*(X_j)$$

Remark.

Thus in practice, we can often assume that the space itself is path connected, while dealing with its homology groups.

Proof. Since $|\Delta_n|$ is a path connected space, it follows that any singular n -simplex in X is contained completely in one of the path components X_j . Therefore, it follows that the entire chain complex $S_*(X)$ is the direct sum of the subchain complexes $S_*(X_j)$. Hence the homology is also a direct sum of $H_*(X_j)$, $j \in J$ by Theorem 5.3.3. $\square$

Theorem 6.2.6 ▶ Homology Long Exact Sequence of the pair (X, A)

By definition of $S_*(X, A)$, we have an exact sequence of chain complexes

$$0 \rightarrow S_*(A) \rightarrow S_*(X) \rightarrow S_*(X, A) \rightarrow 0.$$

By Theorem 6.8.5, this yields the long exact sequence of homology groups

$$\begin{cases} \dots \rightarrow H_i(A) \rightarrow H_i(X) \rightarrow H_i(X, A) \xrightarrow{\delta} H_{i-1}(A) \rightarrow \dots \\ \dots \rightarrow H_1(X, A) \xrightarrow{\delta} H_0(A) \rightarrow H_0(X) \rightarrow H_0(X, A), \end{cases} \quad (4.6)$$

which is functorial.

Remark.

This sequence comes handy in many computations as we shall see soon.

6.3 Homology Excision and Mayer-Vietoris

Lemma 6.3.1

Let $X = A \cup B$. Then the following statements are equivalent.

1. $S_*(A) + S_*(B) \rightarrow S_*(X)$ induces isomorphisms in homology.
2. $[S_*(A) + S_*(B)]/S_*(B) \rightarrow S_*(X)/S_*(B)$ induces isomorphisms in homology.
3. $[S_*(A) + S_*(B)]/S_*(A) \rightarrow S_*(X)/S_*(A)$ induces isomorphisms in homology.
4. $S_*(A)/S_*(A \cap B) \rightarrow S_*(X)/S_*(B)$ induces isomorphisms in homology.
5. $S_*(B)/S_*(A \cap B) \rightarrow S_*(X)/S_*(A)$ induces isomorphisms in homology.

Note.

1,2 等价的好处就在于, 它告诉我们, 我们所需要的同调群的同构对于一部分的信息是不敏感的。

Proof sketch.

对于 1,2 的等价性, 涉及四种同调群, 我们借助 B 的同调将它们相互联系. 用短正合列到长正合列的诱导行为将四种同调群两两联系, 并以 B 的同调为桥梁, 利用诱导行为的函子性将两组同调群联系起来. B 到自身的恒等给出一组同构, 配合

已知的一组同构可以利用 Five lemma 夹出另一组同构.
2,4 的等价性直接就是第二同构定理的结果, 本质上与同调关系不大.

Proof. The equivalence of (1) and (2) is a consequence of the Five lemma applied to the ladder of homology long exact sequences given by the ladder of short exact sequences

$$\begin{array}{ccccccc}
 0 & \longrightarrow & S.(B) & \longrightarrow & S.(A) + S.(B) & \longrightarrow & [S.(A) + S.(B)]/S.(B) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & S.(B) & \longrightarrow & S.(X) & \longrightarrow & S.(X)/S.(B) \longrightarrow 0.
 \end{array}$$

The equivalence of (2) and (4) follows from the commutative diagram below in which the horizontal arrow is the isomorphism given by the Noether's isomorphism theorem.

$$\begin{array}{ccc}
 S.(A)/S.(A \cap B) & \longrightarrow & [S.(A) + S.(B)]/S.(B) \\
 & \searrow & \downarrow \\
 & & S.(X)/S.(B)
 \end{array}$$

□

Definition 6.3.2

Let X be a topological space, A, B be two subspaces such that $X = A \cup B$. We then say that the inclusion map $(A, A \cap B) \hookrightarrow (X, B)$ an excision map. A pair $\{A, B\}$ of subspaces of a space X is called an **excisive couple** for singular homology, if the inclusion map

$$S.(A) + S.(B) \hookrightarrow S.(A \cup B)$$

induces isomorphisms in homology or if any of the other four equivalent conditions in the above lemma is satisfied.

Note.

excisive couple 是一对能代表 (X, A) 和 (X, B) 的同调性质的子集 (A, B) , 是 X 的一个良好分解. 它使得全空间 X 在切除 $X \setminus A$ 或 $X \setminus B$ 的切除行为上, 在同调的意义下是无损的.

Theorem 6.3.3 ▶ Homology excision

The inclusion map $(A, A \cap B) \hookrightarrow (A \cup B, B)$ of an excisive couple $\{A, B\}$ induces an isomorphism of homology groups $H_*(A, A \cap B) \rightarrow H_*(A \cup B, B)$.

Theorem 6.3.4

If $X = X_1 \cup X_2 = \text{int}_X(X_1) \cup \text{int}_X(X_2)$ then, $\{X_1, X_2\}$ is an excisive couple for the singular homology.

Note.

也就是说, 当分解的一个组件的边界能被另一个组件的内部“吃掉”时, 这组分解就是切除对.

Example 6.3.5

Here are a few examples of excisive couples which occur in practice.

1. $\{\mathbb{D}^n, \mathbb{R}^n \setminus \{0\}\}$ is an excisive couple and hence the inclusion map $(\mathbb{D}^n, \mathbb{D}^n \setminus \{0\}) \hookrightarrow (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ will induce isomorphisms in all singular homology groups.
2. Let N denote the north pole on $\mathbb{S}^n$ and $\bar{U}$ denote the closed upper hemisphere. Then $\{\mathbb{S}^n \setminus \{N\}, \bar{U}\}$ is an excisive couple, which gives isomorphisms of $H_*(\bar{U}, \bar{U} \setminus \{N\}) \rightarrow H_*(\mathbb{S}^n, \mathbb{S}^n \setminus \{N\})$.

Theorem 6.3.6

Given subsets X_1 and X_2 of a topological space X , let us denote the inclusion maps of X_i in to X by $\eta_i, i = 1, 2$. They induce inclusion maps of the chain complexes which we shall denote again by $\eta_i : S.(X_i) \rightarrow S.(X), i = 1, 2$. Now consider the chain map

$$(\eta_1, -\eta_2) : S.(X_1) \oplus S.(X_2) \rightarrow S.(X).$$

The submodule generated by $S.(X_1)$ and $S.(X_2)$ in $S.(X)$ is denoted by $S.(X_1) + S.(X_2)$. Then we have a short exact sequence of chain complexes

$$0 \rightarrow S.(X_1 \cap X_2) \rightarrow S.(X_1) \oplus S.(X_2) \rightarrow S.(X_1) + S.(X_2) \rightarrow 0$$

If $\{X_1, X_2\}$ is an excisive couple, then we can obtain the **Mayer-Vietoris**

sequence:

$$\left. \begin{aligned} \cdots \rightarrow H_{i+1}(X_1 \cup X_2) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(X_1 \cup X_2) \rightarrow \cdots \\ \cdots \rightarrow H_1(X_1 \cup X_2) \xrightarrow{\partial} H_0(X_1 \cap X_2) \rightarrow H_0(X_1) \oplus H_0(X_2) \rightarrow H_0(X_1 \cup X_2) \end{aligned} \right\}$$

with

$$i_*(z) = (i_{1*}z, -i_{2*}z); \quad j_*(z_1, z_2) = j_{1*}z_1 + j_{2*}z_2$$

where,

$$\begin{aligned} i_1 : X_1 \cap X_2 \hookrightarrow X_1, \quad i_2 : X_1 \cap X_2 \hookrightarrow X_2 \\ j_1 : X_1 \hookrightarrow X_1 \cup X_2, \quad j_2 : X_2 \hookrightarrow X_1 \cup X_2 \end{aligned}$$

are inclusion maps.

Remark.

The Mayer-Vietoris sequence is one of the major ready-to-use-tools in computing homology.

Proof. The kernel of $(\eta_1, -\eta_2)$ is $S.(X_1) \cap S.(X_2) = S.(X_1 \cap X_2)$. The image is clearly $S.(X_1) + S.(X_2)$. Thus the short sequence is exact. Then we obtain the long exact sequence

$$\begin{aligned} \cdots \rightarrow H_{i+1}(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(S.(X_1) + S.(X_2)) \\ \cdots \rightarrow H_1(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_0(X_1 \cap X_2) \rightarrow H_0(X_1) \oplus H_0(X_2) \rightarrow H_0(S.(X_1) + S.(X_2)) \end{aligned}$$

By the definition of excisive couple, we can replace $H_i(S.(X_1) + S.(X_2))$ by $H_i(X_1 \cup X_2)$ to obtain Mayer-Vietoris. $\square$

6.4 Problems

Problem 6.4.1

Compute the homology of $\mathbb{S}^p \times \mathbb{S}^q$ using Mayer-Vietoris.

Proof. We assume that $p, q \geq 1$. Let U^+, U^- be $\mathbb{S}^q$ without the north pole and south pole respectively. Then $(A, B) := (\mathbb{S}^p \times U^+, \mathbb{S}^p \times U^-)$ is a excisive

couple of $\mathbb{S}^p \times \mathbb{S}^q$. Since $U^+ \cap U^-$ can deformation retract to the equator of $\mathbb{S}^q$, which is $\mathbb{S}^{q-1}$, then $A \cap B$ can deformation retract to $\mathbb{S}^p \times \mathbb{S}^{q-1}$.

$$\cdots \xrightarrow{i'_*} H_{i+1}(\mathbb{S}^p) \oplus H_{i+1}(\mathbb{S}^q) \xrightarrow{j_*} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} H_i(\mathbb{S}^p \times \mathbb{S}^{q-1}) \xrightarrow{i_*} H_i(\mathbb{S}^p) \oplus H_i(\mathbb{S}^q) \cdots$$

The following short exact sequence separates

$$0 \rightarrow \ker \partial \xrightarrow{\iota} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} \text{Im } \partial \rightarrow 0$$

which is equivalent to

$$0 \rightarrow \text{Im } j_* \simeq \text{Coker } j_* \xrightarrow{\iota} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} \ker i_* \rightarrow 0$$

where $\text{Im } j_* \simeq H_{i+1}(\mathbb{S}^p) \oplus H_{i+1}(\mathbb{S}^q) / \text{Im } i'_* = \text{Coker } i'_*$. Thus we have

$$H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \simeq \ker i_* \oplus \text{Coker } i'_*$$

$\text{Im } i_* \simeq H_i$

1. When $p = q$

(a)

□

Exercise 6.4.2

Compute $H_*(\vee_k \mathbb{S}^{n_k})$ where $\vee_k \mathbb{S}^{n_k}$ denotes the bouquet of spheres of varying dimensions n_k . In particular, compute $H_*(\vee_k \mathbb{S}^n)$ where each $n_k = n$.

Proof. Let U be a sufficiently small open neighbourhood near the base point. Let $X_1 = \vee_{k-1} \mathbb{S}^{n_{k-1}} \cup U$. $X_2 = \mathbb{S}^{n_k} \cup U$. X_1 is homotopic equivalent to $\vee_{k-1} \mathbb{S}^{n_{k-1}}$. X_2 is homotopic equivalent to $\mathbb{S}^{n_k}$. $X_1 \cap X_2 = U$ is contractible. The Mayer-Vietoris sequence is

$$\cdots H_{l+1}(\vee_k \mathbb{S}^{n_k}) \xrightarrow{\partial} 0 \rightarrow H_l(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_l(\mathbb{S}^{n_k}) \rightarrow H_l(\vee_k \mathbb{S}^{n_k}) \xrightarrow{\partial} 0 \cdots$$

when $l \geq 2$. Thus

$$H_l(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_l(\mathbb{S}^{n_k}) \simeq H_l(\vee_k \mathbb{S}^{n_k}), \quad l \geq 2$$

By induction , we have

$$H_l(\vee_k \mathbb{S}^{n_k}) \simeq \bigoplus_k H_l(\mathbb{S}^{n_k}), \quad l \geq 2$$

The M-V sequence reduces to

$$0 \rightarrow H_1(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_1(\mathbb{S}^{n_k}) \rightarrow H_1(\vee_k \mathbb{S}^{n_k}) \xrightarrow{\partial} H_0(U) \rightarrow H_0(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_0(\mathbb{S}^{n_k}) \dots$$

when $l = 1$. The exactness show that

$$\text{Im } \partial = \ker(H_0(U) \rightarrow H_0(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_0(\mathbb{S}^{n_k})) = 0$$

Thus

$$0 \rightarrow H_1(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_1(\mathbb{S}^{n_k}) \rightarrow H_1(\vee_k \mathbb{S}^{n_k}) \rightarrow 0$$

And

$$H_1(\vee_k \mathbb{S}^{n_k}) \simeq H_1(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_1(\mathbb{S}^{n_k})$$

By induction

$$H_1(\vee_k \mathbb{S}^{n_k}) \simeq \bigoplus_k H_1(\mathbb{S}^{n_k})$$

And since $\vee_k \mathbb{S}^{n_k}$ is path-connected , finally we have

$$H_l(\vee_k \mathbb{S}^{n_k}) = \begin{cases} \mathbb{Z}, & l = 0 \\ \bigoplus_k H_l(\mathbb{S}^{n_k}), & l \geq 1 \end{cases}$$

Where $H_l(\mathbb{S}^{n_k})$ we have already known. □

6.5 Basic Topology

Problem 6.5.1 ► Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphic to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明: 1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 6.5.2

X is compact, Y is Hausdorff. If $\exists f : X \rightarrow Y$ is a continuous bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$. Since K is compact, $\text{dist}(K, 0) < \infty$. If $x \neq 0$, then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\text{dist}(K, 0) \cdot \|x\|}$. Then $\left\|\frac{x}{\varepsilon_0}\right\| > \text{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction. For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

, which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx , we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle inequality, we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex, we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta} \left((N(x_2) + \delta) \frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta) \frac{x_1}{N(x_1) + \delta} \right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \text{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x) \frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continuous, only need to show it is continuous at 0. In fact, $\|\lim_{x \rightarrow 0} \varphi(x)\| = \lim_{x \rightarrow 0} \|\varphi(x)\| = \lim_{x \rightarrow 0} N(x) = 0$ since N is a norm. We define $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)} x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \text{Id}$. The same skills can be used to show ψ is continuous at 0 as well. Finally, by the lemma above, X, Y are homeomorphic.

□

Problem 6.5.3 ► Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n -dimensional projective space $\mathbb{K}\mathbb{P}^n$ is the orbit space of $\mathbb{K}^{n+1} \setminus \{0\}$ under the action of $\mathbb{K}^*$ by rescaling.

1. Prove that $\mathbb{R}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \setminus \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{C}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$,

where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.

3. Prove that $\mathbb{RP}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{CP}^1$ is homeomorphic to $\mathbb{S}^2$.

Proof. 1. Define

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n, \quad \pi(x) = [x]$$

and define

$$f : \mathbb{S}^n \rightarrow \mathbb{RP}^n = \pi|_{\mathbb{S}^n}$$

Then f is continuous. Take any $p \in \mathbb{RP}^n$, there exists $v \in \mathbb{R}^{n+1} \setminus \{0\}$, such that $p = [v]$. Let $x = \frac{v}{\|v\|}$, then $x \in \mathbb{S}^n$,

$$f(x) = [x] = [v] = p$$

thus f is surjective. If

$$f(x) = f(y)$$

, then $[x] = [y]$, there exists $\lambda \in \mathbb{R}^*$, such that $x = \lambda y$. Since

$$\|x\| = \|\lambda y\| = |\lambda| \|y\| \implies |\lambda| = 1$$

, $\lambda = \pm 1$, $x = y$ or $x = -y$. Thus $\text{Fiber}_p(f) = \text{Orbit}_x(\{\pm 1\})$. The universal property of quotient topology gives a continuous bijection

$$\tilde{f} : \mathbb{S}^n \setminus \{\pm 1\} \rightarrow \mathbb{RP}^n$$

Since $\mathbb{S}^n \setminus \{\pm 1\}$ is compact, $\mathbb{RP}^n$ is Hausdorff, $\tilde{f}$ is a homeomorphism.

2. Similarly, let $g : \mathbb{S}^{2n+1} \rightarrow \mathbb{CP}^n$, $g(z) = [z]$. Take any $p = [w] \in \mathbb{CP}^n$, where $w \in \mathbb{C}^{n+1} \setminus \{0\}$. Let $z = \frac{w}{\|w\|}$. Then

$$g(z) = [z] = [w] = p$$

g is surjective. If

$$g(z_1) = g(z_2)$$

, then there exists $\lambda \in \mathbb{C}^*$, such that $z_1 = \lambda z_2$, then

$$\|z_1\| = |\lambda| \|z_2\| \implies |\lambda| = 1$$

which gives that $\lambda \in \mathbb{S}^1$. Thus $\text{Fiber}_p(f) \simeq \text{Orbit}_x(\mathbb{S}^1)$. Thus

$$\mathbb{CP}^n \simeq \mathbb{S}^{2n+1}/\mathbb{S}^1$$

3. (a) Only need to show that $\mathbb{S}^1/\{\pm 1\} \simeq \mathbb{S}^1$,

$$h(z) = z^2$$

induces a continuous bijection $\tilde{h} : \mathbb{S}^1 \setminus \{\pm 1\} \rightarrow \mathbb{S}^1$. $\mathbb{S}^1 \setminus \{\pm 1\}$ is compact, $\mathbb{S}^1$ is Hausdorff, $\tilde{h}$ is a homeomorphism.

(b) Since $\mathbb{S}^3/\mathbb{S}^1 \simeq \mathbb{S}^2$.

□

Problem 6.5.4

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$.
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translations.

Proof. 1. Let

$$f(x, y) := (e^{2\pi xi}, e^{2\pi yi})$$

Then

$$f(x, 0) = (e^{2\pi ix}, 1) = f(x, 1)$$

$$f(0, y) = (1, e^{2\pi yi}) = f(1, y)$$

Thus f induces a map $\bar{f}$ on $\mathbb{T}$ such that $\bar{f}([(x, y)]) = (e^{2\pi xi}, e^{2\pi yi})$, which is a continuous function. It is not hard to see that $\bar{f} : \mathbb{T} \rightarrow \mathbb{S}^1 \times \mathbb{S}^1$ is a bijection. As a continuous image of $[0, 1]^2$, $\mathbb{T}$ is compact. Take any closed subset V of $\mathbb{T}$, V is compact. $\bar{f}(V)$ is compact as the continuous image of V . Since $\mathbb{S}^1 \times \mathbb{S}^1$ is Hausdorff, $\bar{f}(V)$ is closed. Thus $\bar{f}$ is a continuous and closed bijection, thus is a homeomorphism.

2. Let

$$f(x, y) = (x, y)/\mathbb{Z}^2$$

Then Since $(x, 0) - (x, 1) \in \mathbb{Z}^2$, $(0, y) - (1, y) \in \mathbb{Z}^2$. f induces a continuous map $\bar{f}$ on from $\mathbb{T}$ to $\mathbb{S}^1 \times \mathbb{S}^1$. It is clear that f maps onto $\mathbb{R}^2/\mathbb{Z}^2$. If

$f(x_1, y_1) = f(x_2, y_2)$, then

$$(x_1 - x_2, y_1 - y_2) \in \mathbb{Z}^2 \implies x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\}, \text{ and } y_1 = y_2 \text{ or } \{y_1, y_2\} = \{0, 1\}$$

thus $(x_1, y_1) \sim (x_2, y_2)$. Again since $\mathbb{T}$ is compact, $\mathbb{R}^2/\mathbb{Z}^2$ is Hausdorff, $\bar{f}$ is a homomorphism. □

Problem 6.5.5

Consider the orthogonal group $O_{n-1}(\mathbb{R})$ as the subgroup of $O_n(\mathbb{R})$ of matrices of the form:

$$\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in O_n(\mathbb{R}), A \in O_{n-1}(\mathbb{R}).$$

Let $O_{n-1}(\mathbb{R})$ act on $O_n(\mathbb{R})$ by left multiplication. Prove that the orbit space $O_n(\mathbb{R})/O_{n-1}(\mathbb{R})$ is homeomorphic to $\mathbb{S}^{n-1}$.

Proof sketch.

将 $O_n(\mathbb{R})$ 刻画为 $\mathbb{S}^{n-1}$ 上的群作用. 此时 $A \sim \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}$ 就是北极点的稳定子群. 一但作用是传递的, 由流形上的轨道-稳定化子定理, 就有

$$O_n(\mathbb{R})/O_{n-1}(\mathbb{R}) \simeq \mathbb{S}^{n-1}$$

6.6 Homotopy

Problem 6.6.1 ▶ Möbius band

Let M be the Möbius band, that is, the quotient of $[0, 1]^2$ under the equivalence relation generated by $(x, 0) \sim (1 - x, 1)$ for all $x \in [0, 1]$. Prove that M is homotopy equivalent to $\mathbb{S}^1$.

Proof. Define a map

$$H : [0, 1]^2 \times [0, 1] \rightarrow \mathbb{S}^1, \quad H((x, y), t) = \left((1 - t)x + \frac{1}{2}t, y \right)$$

Let $\tilde{H} = \pi \circ H$. Then

$$\tilde{H}((x, 0), t) = \pi\left((1-t)x + \frac{1}{2}t, 0\right)$$

$$\tilde{H}((1-x, 1), t) = \pi\left(\left((1-t)(1-x) + \frac{1}{2}t\right), 1\right)$$

Since

$$1 - \left((1-t)x + \frac{1}{2}t\right) = (1-t)(1-x) + \frac{1}{2}t$$

,

$$\tilde{H}((x, 0), t) = \tilde{H}((1-x, 1), t)$$

$\tilde{H}$ induces a continuous map H^* on M . And

$$H^*([(x, y), 0]) = [(x, y)], \quad H^*([(x, y), 1]) = \left[\left(\frac{1}{2}, y\right)\right]$$

Thus H^* is a deformation retract from M to $\mathbb{S}^1$, which is a homotopy equivalence as well. $\square$

Problem 6.6.2

Let X be space. The cone CX is the quotient space:

$$CX = (X \times [0, 1]) / (X \times \{0\}).$$

Let $\pi : X \times [0, 1] \rightarrow CX$ be the quotient map and $\iota : X \rightarrow CX, x \mapsto \pi(x, 1)$.

1. Let $f : X \rightarrow Y$ be a map. Prove that f is homotopic to a constant map if and only if there exists $f' : CX \rightarrow Y$ such that $f' \circ \iota = f$.
2. Prove that CS^n is homeomorphic to D^{n+1} .
3. Prove that for all X , CX is contractible.

Proof. 1. If such f' exists. Then $H := f' \circ \pi : X \times [0, 1] \rightarrow Y$ is a homotopy from f to the constant map valued the vertex of CX . Conversely, if f is homotopic to constant map. Suppose $H : X \times I \rightarrow Y$ is its homotopy map. Since CX is the quotient space of $X \times [0, 1]$, and $H(X \times \{0\}) = \text{constant}$. It follows from the universal property that H induces a map $f' : CX \rightarrow Y$ such that $f' \circ \pi = H$. Finally

$$(f' \circ \iota)(x) = f'(\pi(x, 1)) = H(x, 1) = f(x)$$

2. Define

$$g : \mathbb{S}^n \times [0, 1] \rightarrow D^{n+1}, \quad g(x, t) = t \cdot x$$

Since $g(x, 0) \equiv 0 \in D^{n+1}$, g induces a continuous function $h : CS^n \rightarrow D^{n+1}$ such that $g \circ \pi = h$. We now show that h is a bijection. If

$$h([x_1, t_1]) = h([x_2, t_2])$$

, that is $t_1 x_1 = t_2 x_2$, then

$$\|t_1 x_1\| = \|t_2 x_2\| \implies t_1 = t_2$$

If $t_1 = t_2 = 0$, then $[x_1, t_1] = [x_2, t_2] = 0$. If $t_1 = t_2 > 0$, then $x_1 = x_2$, thus $[x_1, t_1] = [x_2, t_2]$. Thus h is injective. For surjectivity, let $y \in D^{n+1}$. If $y = 0$, we take $t = 0$, then $h([x, 0] = 0) = y$. If $y \neq 0$, then

$$h\left(\left[\frac{y}{\|y\|}, \|y\|\right]\right) = y$$

Thus h is surjective. Since CS^n is compact, D^{n+1} is Hausdorff, h is homeomorphism.

3. Define

$$H : CX \times [0, 1] \rightarrow CX, \quad H([x, s], t) = [x, s \cdot (1 - t)]$$

Then it is not hard to see that H is well-defined and continuous. Since

$$H([x, s], 0) = [x, s], \quad H([x, s], 1) = [x, 0]$$

H is a homotopy from Id to Constant.

□

Problem 6.6.3 ► Linear Groups

1. Prove that the inclusion $O_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is a homotopy equivalence. (Use the Gram-Schmidt orthonormalization procedure to construct the reverse map).
2. Among the following matrix groups, determine which ones are compact, and determine their π_0 : $GL_n(\mathbb{C})$, $GL_n(\mathbb{R})$, $O_n(\mathbb{R})$, $SO_n(\mathbb{R})$, $U_n(\mathbb{C})$, $SU_n(\mathbb{C})$.

Proof. 1. Take $A \in GL_n(\mathbb{R})$, suppose $A = (v_1, \dots, v_n)$. Then $\{v_1, \dots, v_n\}$ passing to an orthonormal basis $\{u_1, \dots, u_n\}$ by Gram-Schmidt orthonor-

malization. We write $r(A) = \{u_1, \dots, u_n\}$. Then r is a continuous map on $GL_n(\mathbb{R})$ since the inner product is continuous. We define

$$H : GL_n(\mathbb{R}) \times [0, 1] \rightarrow GL_n(\mathbb{R}), \quad H(A, t) = (1 - t)A + tr(A)$$

we need to verify that $H(A, t) \in GL_n(\mathbb{R})$ for all A, t . The k -th column vector of $H(A, t)$ is $(1 - t)v_k + tu_k$. Since

$$v_k \in \text{span}\{u_1, \dots, u_{k-1}\}$$

we can write

$$(1 - t)v_k + tu_k = (1 - t) \sum_{j=1}^k c_{kj}u_j + tu_k$$

Since $c_{jj} > 0$, $(1 - t)c_{jj} + t > 0$, $\{(1 - t)v_1 + tu_1, \dots, (1 - t)v_k + tu_k\}$ are linealy independent. And since

$$H(A, 0) = A = \text{Id}_{GL_n(\mathbb{R})}(A)$$

$$H(A, 1) = r(A) \in O_n(\mathbb{R})$$

and for $B \in O_n(\mathbb{R})$, $H(B, t) = r(B) = 1$. We have $H : \text{Id}_{GL_n(\mathbb{R})} \simeq i$ is a SDR.

2. $GL_n(\mathbb{C})$ is a open submanifold of $M_n(\mathbb{C})$, which is noncompact. Since $GL_n(\mathbb{C})$ is path-connected. $\pi_0(GL_n(\mathbb{C})) = 1$. $GL_n(\mathbb{R})$ is a open submanifold of $M_n(\mathbb{R})$, thus $GL_n(\mathbb{R})$ is noncompact. $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$ is a continuous surjection. Since $\mathbb{R} \setminus \{0\}$ has two components. $GL_n(\mathbb{R})$ has at least two components. Since

$$GL_n^+(\mathbb{R}) = \{A : \det A > 0\}, \quad GL_n^-(\mathbb{R}) = \{A : \det A < 0\}$$

are exactly two components of $GL_n(\mathbb{R})$. $\pi_0(GL_n(\mathbb{R})) = 2$. Since $\{A : A^T A = E\}$ is closed. $O_n(\mathbb{R})$ is bounded. $O_n(\mathbb{R})$ is compact. $\pi_0(O_n(\mathbb{R})) = 2$. $SO_n(\mathbb{R})$ is a closed subset of $O_n(\mathbb{R})$, thus is compact as well. $\pi_0(SO_n(\mathbb{R})) = 1$. $U_n(\mathbb{C})$ is a closed subset of $GL_n(\mathbb{C})$. $\pi_0(U_n(\mathbb{C})) = 1$. $SU_n(\mathbb{C})$ is compact, $\pi_0(SU_n(\mathbb{C})) = 1$.

□

Problem 6.6.4

1. Prove that $\mathbb{CP}^{n+1}$ is obtained from $\mathbb{CP}^n$ by gluing a cell of dimension $2n + 2$.
2. Compute the fundamental group of $\mathbb{CP}^n$ for $n \geq 1$.

6.7 van Kampen

Lemma 6.7.1

设 X 分解为一些道路连通集 A_α 的并, 且每个 A_α 包含了基点 $x_0 \in X$. 则所有含入映射 $A_\alpha \hookrightarrow X$ 的诱导同态 $j_\alpha : \pi_1(A_\alpha) \rightarrow \pi_1(X)$ 扩张为同态 $\Phi : *_\alpha \pi_1(A_\alpha) \rightarrow \pi_1(X)$.

Proposition 6.7.2

令 $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(A_\alpha)$ 表示含入映射 $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$ 诱导的同态, 则 $j_\alpha i_{\alpha\beta} = j_\beta i_{\beta\alpha}$. 所以 $\ker \Phi$ 包含所有形如 $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$, $\omega \in \pi_1(A_\alpha \cap A_\beta)$ 的元素.

Note.

即 Φ 将 $A_\alpha \cap A_\beta$ 落在 A_α 和 A_β 中部分的像等同.

Theorem 6.7.3 ▶ van Kampen

设 X 分解为包含了基点 $x_0 \in X$ 的道路连通开集 A_α 的并. 且每个交集 $A_\alpha \cap A_\beta$ 是道路连通的, 则同态 $\Phi : *_\alpha \pi_1(A_\alpha) \rightarrow \pi_1(X)$ 是满射. 此外, 若每个 $A_\alpha \cap A_\beta \cap A_\gamma$ 也是道路连通的, 则 Φ 的核 N 由全体形如 $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$, $\omega \in \pi_1(A_\alpha \cap A_\beta)$ 的元素生成, 从而 Φ 诱导出同构

$$\pi_1(X) \simeq *_\alpha \pi_1(A_\alpha) / N = *_\alpha \pi_1(A_\alpha) / (i_{\alpha\beta}(\omega) = i_{\beta\alpha}(\omega))$$

Example 6.7.4

设 $\{X_\alpha\}$ 是一族道路连通的空间, $x_\alpha \in X_\alpha$ 是基点, 考虑它们的楔和 $\bigvee_\alpha X_\alpha$. 对于每个 x_α , 它在 X_α 中都存在形变收缩到自身的开邻域 U_α . 令 $A_\alpha = X_\alpha \bigvee_{\beta \neq \alpha} U_\beta$, 则 A_α 形变收缩到 X_α . 此外, $\bigcap_\alpha A_\alpha$ 形变收缩到点 x_α . 由 van Kampen 定理, $\Phi : \pi_1(X) \rightarrow *_\alpha \pi_1(A_\alpha) \simeq *_\alpha \pi_1(X_\alpha)$ 是同构.

6.8 Homology

Lemma 6.8.1 ▶ Snake lemma

Given a commutative diagram of R -module homomorphisms: where the two horizontal sequences are exact, there exists a R -module homomorphism

$$\begin{array}{ccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\alpha'} & N & \xrightarrow{\beta'} & N'' \end{array}$$

$\delta : \text{Ker } f'' \longrightarrow \text{Coker } f'$, called the *connecting homomorphism* such that the sequence

$$\text{Ker } f' \xrightarrow{\bar{\alpha}} \text{Ker } f \xrightarrow{\bar{\beta}} \text{Ker } f'' \xrightarrow{\delta} \text{Coker } f' \xrightarrow{\bar{\alpha}'} \text{Coker } f \xrightarrow{\bar{\beta}'} \text{Coker } f''$$

is exact. Moreover, the *connecting homomorphism* δ has the naturality properties, so that the above assignment of a ‘snake’ to the corresponding ‘six-term’ exact sequence of modules defines a covariant functor.

Note.

最关键的信息是 *connecting homomorphism* δ . 构造的过程就是借助 β 的是 surjective, α' 是 injective, 来调整使得 β 和 α' 对应的箭头是可以反转的. 事实上在应用时, 这个 connecting homomorphism 的具体构造往往比它的存在性更重要.

Proof. to be finished We need to show

1. $\bar{\alpha}, \bar{\beta}$ are well defined, where $\bar{\alpha} := \alpha|_{\text{Ker } f'}, \bar{\beta} := \beta|_{\text{Ker } f}$
2. $\text{Im } \bar{\alpha} = \text{Ker } \bar{\beta}$.
3. $\bar{\alpha}', \bar{\beta}'$ are well defined, where $\bar{\alpha}'([x]) := [\alpha'(x)], \bar{\beta}'([y]) := [\beta'(y)]$
4. $\text{Im } \bar{\alpha}' = \text{Ker } \bar{\beta}'$
5. Construct a δ .
6. $\text{Im } \bar{\beta} = \text{Ker } \delta$

7. $\text{Im } \delta = \ker \bar{\alpha}'$

We prove the above points below.

1. By commutative, $f \circ \alpha(\ker f') = \alpha' \circ f'(\ker f') = 0$, which shows that $\alpha(\ker f') \subseteq \ker f$. Thus $\bar{\alpha}$ is well defined. $\bar{\beta}$'s is similar.

- 2.

$$\text{Im } \bar{\alpha} = \alpha(\ker f') \subseteq \text{Im } \alpha = \ker \beta \implies \beta(\text{Im } \bar{\alpha}) = 0 \implies \text{Im } \bar{\alpha} \subseteq \ker \bar{\beta}$$

The other side,

$$\ker \bar{\beta} = \ker \beta \cap \ker f = \text{Im } \alpha \cap \ker f$$

Take $y \in \ker \bar{\beta}$, there exists $x \in M'$ such that $y = \alpha(x)$ and $f(y) = f(\alpha(x)) = 0$. Since $f \circ \alpha = \alpha' \circ f'$, $\alpha' \circ f'(x) = 0$. Since α' is injective, $x \in \ker f'$. Thus $y \in \text{Im } \bar{\alpha}$.

3. For $\bar{\alpha}'$, we need to show that for $x_1, x_2 \in N'$ such that $x_1 - x_2 \in \text{Im } f'$, $\alpha'(x_1) - \alpha'(x_2) \in \text{Im } f$. It is true by commutative

$$\alpha'(x_1) - \alpha'(x_2) = \alpha'(x_1 - x_2) \in \text{Im } (\alpha' \circ f') = \text{Im } (f \circ \alpha) \subseteq \text{Im } f$$

$\bar{\beta}'$'s is similar.

4. Take $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, then there exists $[x]_{\sim \text{Im } f'} \in \text{Coker } f'$, such that $\alpha'([x]_{\sim \text{Im } f'}) = [y]_{\sim \text{Im } f}$. Then $\alpha'(x) - y \in \text{Im } f$, there exists $z \in M$, such that

$$\alpha'(x) - y = f(z)$$

By commutative and exactness: $\ker \beta' = \text{Im } \alpha'$,

$$\beta'(y) = \beta'(\alpha'(x) - f(z)) = 0 - \beta' \circ f(z) = -f'' \circ \beta(z) \in \text{Im } f''$$

, which shows that $\bar{\beta}'([y]_{\sim \text{Im } f}) = [0]_{\sim \text{Im } f''}$, $\text{Im } \bar{\alpha}' \subseteq \ker \bar{\beta}'$.

Then, take $[y]_{\sim \text{Im } f} \in \ker \bar{\beta}'$. Then $y \in \ker \beta' = \text{Im } \alpha'$. There exists $x \in N'$ such that $y = \alpha'(x)$. Then $[y]_{\sim \text{Im } f} = \alpha'([x]_{\sim \text{Im } f'})$, $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, $\ker \bar{\beta}' \subseteq \text{Im } \bar{\alpha}'$.

5. Take $z \in \ker f''$. Since β is surjective, there exists $y \in M$ such that $\beta(y) = z$. Once we show that $f(y) \in \text{Im } \alpha'$, we can define

$$\delta(z) = [(\alpha')^{-1}(f(y))]_{\sim \text{Im } f'}$$

provided that this definition is well defined. By commutative

$$\beta'(f(y)) = f''(\beta(y)) = f''(z) = 0.$$

Hence $f(y) \in \ker \beta' = \text{Im } \alpha'$. Finally, to show that it is well defined, we take $z_1, z_2 \in \ker f''$ such that $\beta(y_1) = \beta(y_2) = z$. We need to show that $(\alpha')^{-1}(f(y_1 - y_2)) \in \text{Im } f'$. Since $\beta(y_1) - \beta(y_2) = 0$, $\beta(y_1 - y_2) = 0$, $y_1 - y_2 \in \ker \beta = \text{Im } \alpha$. Suppose $\alpha(x) = y_1 - y_2$. Then

$$(\alpha')^{-1}(f(y_1 - y_2)) = (\alpha')^{-1}(f\alpha(x)) = (\alpha')^{-1}(\alpha' \circ f')(x) = f'(x) \in \text{Im } f'$$

, which completes the proof. □

Corollary 6.8.2

Consider the following commutative diagram of R -modules and R -linear maps in which the two rows are exact. If f_1 and f_3 are isomorphisms then so is f_2 .

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \longrightarrow & 0 \\ & & \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \\ 0 & \longrightarrow & N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \longrightarrow & 0 \end{array}$$

Proof. By snake lemma, we have

$$0 \xrightarrow{\alpha_1} \ker f_2 \xrightarrow{\alpha_2} 0 \xrightarrow{\delta} 0 \xrightarrow{\beta_1} \text{Coker } f_2 \xrightarrow{\beta_2} 0$$

is exact. Then we have

$$\begin{aligned} 0 &= \ker \alpha_2 = \text{Im } \alpha_1 = 0 \\ &= \ker f_2 \end{aligned}$$

which shows that $\ker f_2 = 0$. Similarly, $\text{Coker } f_2 = 0$. The above shows that f_2 is an isomorphism. □

Corollary 6.8.3 ▶ Four lemma

Consider the following commutative diagram of R modules and R -linear maps in which the two rows are exact. Suppose that f_1 is surjective and

f_4 is injective. Then

(i) f_2 is injective $\implies f_3$ is injective.

(ii) f_3 is surjective $\implies f_2$ is surjective.

$$\begin{array}{ccccccc} M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & M_4 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 \\ N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 \end{array}$$

Proof sketch.

如果 f_3 被两个单射 f_2, f_4 夹在中间, 得益于 f_1 是满射, 可以将右三列保持单射性地调整为一条 snake, 从而导出 f_3 是单射.

类似地, 如果 f_2 被两个满射 f_1, f_3 夹在中间, 得益于 f_4 是单射, 左三列可以保持满射性地调整为一条 snake, 导出 f_2 是满射.

Proof. 1. We can construct a snake

$$\begin{array}{ccccccc} M_2/\ker \alpha_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & \operatorname{Im} \alpha_3 & \longrightarrow & 0 \\ \downarrow \bar{f}_2 & & \downarrow f_3 & & \downarrow f_4|_{\operatorname{Im} \alpha_3} & & \\ 0 \longrightarrow & N_2/\ker \beta_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 & \end{array}$$

By snake lemma, the following sequence is exact

$$\ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3}$$

Since f_4 is injective, $f_4|_{\operatorname{Im} \alpha_3}$ is as well. Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc} M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{p} & M_2/\ker \alpha_2 & \longrightarrow & 0 \\ \bar{f}_1 \downarrow & & \downarrow f_2 & & \downarrow \bar{f}_2 & & \\ 0 \longrightarrow & N_1/\ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{p} & N_2/\ker \beta_2 & \end{array}$$

We get

$$0 = \ker f_2 \rightarrow \ker \bar{f}_2 \xrightarrow{\delta} \operatorname{Coker} \bar{f}_1$$

is a exact sequence, where $\operatorname{Coker} \bar{f}_1 = 0$ since f_1 is surjective and $\bar{f}_1$ is as

well. It shows that $\ker \bar{f}_2 = 0$. Finally, we have the following sequence is exact

$$0 = \ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\text{Im } \alpha_3} = 0$$

It follows that $\ker f_3 = 0$, f_3 is injective.

2. Another snake we can construct is

$$\begin{array}{ccccccc} M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & \text{Im } \alpha_2 = \ker \alpha_3 & \longrightarrow & 0 \\ \downarrow \bar{f}_1 & & \downarrow f_2 & & \downarrow f_3|_{\text{Im } \alpha_2} & & \\ 0 & \longrightarrow & N_1 / \ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{\beta_2} & \text{Im } \beta_2 = \ker \beta_3 \end{array}$$

Then by snake lemma, the following sequence is exact

$$\ker \bar{f}_1 \rightarrow \ker f_2 \rightarrow \ker f_3|_{\text{Im } \alpha_2} \rightarrow \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2}$$

Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ker } \alpha_3 & \xrightarrow{\iota_M} & M_3 & \xrightarrow{\alpha_3} & \text{Im } \alpha_3 \longrightarrow 0 \\ & & \downarrow f_3|_{\text{Im } \alpha_2} & & \downarrow f_3 & & \downarrow f_4|_{\text{Im } \alpha_3} \\ 0 & \longrightarrow & \text{Ker } \beta_3 & \xrightarrow{\iota_N} & N_3 & \xrightarrow{\beta_3} & \text{Im } \beta_3 \longrightarrow 0 \end{array}$$

We get

$$0 = \ker f_4|_{\text{Im } \alpha_3} \xrightarrow{\delta} \text{Coker } (f_3|_{\text{Im } \alpha_2}) \rightarrow \text{Coker } f_3 = 0$$

which shows that $\text{Coker } (f_3|_{\text{Im } \alpha_2}) = 0$. Finally, we have the following sequence exact

$$0 = \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2} = 0$$

Thus $\text{Coker } f_2 = 0$, f_2 is surjective. □

Corollary 6.8.4 ► Five lemma

In the following diagram of R -modules, the two rows are given to be exact. If f_1, f_2, f_4 and f_5 are isomorphisms then f_3 is also an isomorphism.

$$\begin{array}{ccccccccc}
M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & M_4 & \longrightarrow & M_5 \\
\downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\
M'_1 & \longrightarrow & M'_2 & \longrightarrow & M'_3 & \longrightarrow & M'_4 & \longrightarrow & M'_5
\end{array}$$

Proof. By applying the Four lemma to the left four columns, we get f_3 is injective. And by applying the Four lemma to the right four columns, we get f_3 is surjective.

□

Theorem 6.8.5

Given a short exact sequence of chain complexes

$$0 \rightarrow C'_* \xrightarrow{\alpha} C_* \xrightarrow{\beta} C''_* \rightarrow 0$$

there is a functorial long exact sequence of homology groups

$$\rightarrow H_n(C'_*) \xrightarrow{H_n(\alpha)} H_n(C_*) \xrightarrow{H_n(\beta)} H_n(C''_*) \xrightarrow{\delta_n} H_{n-1}(C'_*) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(C_*) \rightarrow$$

Proof sketch.

Consider the diagram

$$\begin{array}{ccccccc}
C'_n/\text{Im}\partial'_{n+1} & \xrightarrow{\bar{\alpha}_n} & C_n/\text{Im}\partial_{n+1} & \xrightarrow{\bar{\beta}_n} & C''_n/\text{Im}\partial''_{n+1} & \longrightarrow & 0 \\
\downarrow \partial'_n & & \downarrow \partial_n & & \downarrow \partial''_{n+1} & & \\
0 \longrightarrow & \text{Ker}\partial'_{n-1} & \xrightarrow{\alpha'_{n-1}} & \text{Ker}\partial_{n-1} & \xrightarrow{\beta'_{n-1}} & \text{Ker}\partial''_{n-1} &
\end{array}$$

Example 6.8.6 ► reduced homology

定义 ε 为将 0-链映为系数和的同态. 称同态 ε 为增广同态, 可以按以下方式扩张

出链复形 $\tilde{S}(X)$

$$\tilde{S}_n(X) = \begin{cases} S_n(X), & \text{if } n \geq 0, \\ \mathbb{Z}, & \text{if } n = -1, \\ (0), & \text{if } n < -1; \end{cases} \quad \text{and} \quad \tilde{\partial}_n = \begin{cases} \partial_n, & n \geq 1, \\ \varepsilon, & n = 0, \\ 0, & n < 0. \end{cases}$$

相应的同调群记作 $\tilde{H}_*(X)$, 称为约化同调群. 显然 $\tilde{H}_i(X) = (0), i < 0$; $\tilde{H}_i(X) \simeq H_i(X), i > 1$; $\tilde{H}_0(X) \oplus \mathbb{Z} \simeq H_0(X)$. 特别地, 对于道路连通空间 X , $\tilde{H}_0(X) = 0$.

Problem 6.8.7 ► cofibration

Let (X, A) be a pair of spaces. We say that the inclusion $A \rightarrow X$ is a cofibration if, whenever $f : X \rightarrow Y$ and $H : A \times [0, 1] \rightarrow Y$ are maps such that $H(a, 0) = f(a)$ for all $a \in A$, there exists $\tilde{H} : X \times [0, 1] \rightarrow Y$ such that $\tilde{H}(a, t) = H(a, t)$ for all $(a, t) \in A \times [0, 1]$ and $\tilde{H}(x, 0) = f(x)$ for all $x \in X$.

1. Prove that the inclusion $(X, A) \rightarrow (X \cup CA, CA)$ induces an isomorphism on relative homology.
2. Prove that if $A \rightarrow X$ is a cofibration and A is contractible, then the quotient map $X \rightarrow X/A$ is a homotopy equivalence.
3. Prove that if X is obtained from A by gluing cells, then $A \rightarrow X$ is a cofibration.
4. Prove that if $A \rightarrow X$ is a cofibration, then $(X \cup CA, CA)$ is a cofibration.
5. Let $A \rightarrow X$ be a cofibration. Prove that the quotient map induces an isomorphism $H_i(X, A) \cong H_i(X/A, A/A) = \tilde{H}_i(X/A)$.

Problem 6.8.8 ► homology of suspension

Let X be a space. Compute $H_*(\Sigma X)$ in terms of $H_*(X)$.

Problem 6.8.9

Compute the homology of $\mathbb{CP}^n$.

6.9 Degree

Problem 6.9.1

Let $f : S^n \rightarrow S^n$ be a map. Its **degree** $\deg(f) \in \mathbb{Z}$ is the integer such that for all $x \in H_n(S^n; \mathbb{Z})$, we have $f_*(x) = \deg(f) \cdot x$.

1. Prove that this definition matches with the definition with fundamental groups for $n = 1$.
2. Prove that if $\deg(f) \neq 0$ then f is surjective. Find a counterexample for the converse.
3. Prove that if f is injective, then $\deg(f) = \pm 1$. Find a counterexample for the converse.
4. Prove that the degree of a reflection is -1 (use Mayer-Vietoris). Given $A \in O_n(\mathbb{R})$, prove that the degree of $x \mapsto Ax$ is equal to $\det(A)$. Compute the degree of the antipodal map $x \mapsto -x$.
5. Prove that ΣS^n is homeomorphic to S^{n+1} , then prove that the degree of f is equal to the degree of its suspension $\Sigma f : S^{n+1} \rightarrow S^{n+1}$.

Problem 6.9.2

1. Prove that any map $f : S^n \rightarrow S^n$ without fixpoints is homotopic to the antipodal map. (Use that for all $x \in S^n$, the origin $0 \in \mathbb{R}^{n+1}$ is not on the segment $[f(x), -x]$).
2. Prove that if n is even, then any $f : S^n \rightarrow S^n$ admits a fixpoint.
3. Prove that if a group G acts freely on S^n with n even, then $\#G \leq 2$.
4. Prove that S^n , for n odd, admits a nonvanishing vector field.
5. Prove that any vector field on S^n , for n even, vanishes at some point. (Given such a vector field X , consider the map $u \mapsto X(u)/\|X(u)\|$).

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